



EMGNET
Classification of Limb Movements using Surface EMG Signals

Workshop on advances in deep learning and applications.
WADLA 2021, IIIT sri city, Chittoor, AP

Team & Sponsorship

- Dr. Anish Chand Turlapaty
  - Signal Processing and Pattern Recognition
- Dr. Shiv Ram Dubey (Collaborator)
  - Deep learning
- Dr. Balakrishna Gokaraju (Collaborator)
  - Visualizations and Computing Advanced Research Center (ViCAR) & Computational Science and Engineering (CSE), NCAT State University, Greensboro, North Carolina, USA.
  - Machine Learning & Robotics
- PhD Student: Mr. Naveen Kumar
- SERB – CORE RESEARCH GRANT - **CRG/2019/003801**
  - **Feb 2020 – Feb 2022**

Outline

- MOTIVATION
- ANATOMY AND PHYSIOLOGY
- FUNDAMENTALS OF SURFACE EMG SIGNALS
- DATASETS
- CLASSIFICATION
  - CLASSICAL APPROACH
  - DEEP LEARNING APPROACH
- DEMONSTRATION

MOTIVATION

Upper Limb Disabilities - REHAB






- Leading Cause: Stroke to important regions in the Brain
- Disabilities: Motor impairments for moving the limb
- Recovery: Rehabilitation can help in functional recovery
- Challenge: Rehabilitation is both physically and mentally intensive
- Variations in the extent of disability

- Main Goal: To provide limb mobility assistance to a patient to enable faster recovery
- Impact: Enables the patient to carryout the therapeutic movement patterns for an extended duration
- Care Provider: Can spend more time on design and adaptation of the therapy
- Challenge: To improve the exoskeleton system design to mimic the skills of a human care provider

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Upper Limb Exoskeletons

- Armeo power



Company: Hocoma,
a German MNC working on human movement therapy
Application: Used in a rehab center to help injured/disabled people
DoF: 7

- Fourier m1



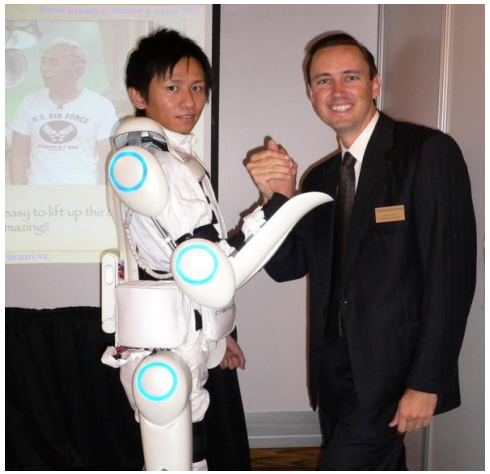
Inventor: Fourier Intelligence, a Chinese company working on limb rehab therapy
Application: Used in a rehab center to help injured/disabled people
Principle: The Brunnstrom approach

m-power Arm Brace



Inventors: myomo, my own motion, grown at MIT, Oxford, MA
Application: Function restoration of upper limbs for people with Brachial plexus injury
Tech: Myoelectric signal based

Hybrid Assistive Limb (HAL)



Inventors: Tsukuba University and Cyberdyne.
Application: HAL 3 is full body exoskeleton
Tech: Based on Myoelectric signals

Desired Characteristics

- Complete wearability
- Adaptability
- Portability
- Energy Smart
- Reliability of the device
- User's safety

Major Challenges for HRI

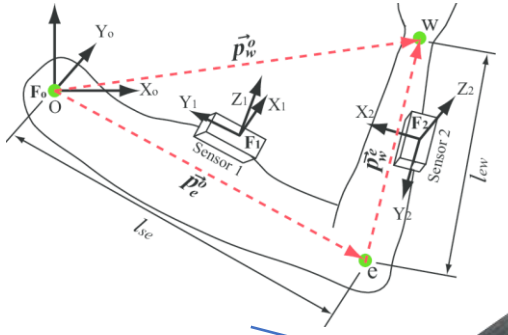
- To implement degrees of freedom that mimic the natural human movements
  - Learn and adapt to human kinematics
  - Proprioception: accelerometers, encoders, and tachometers
  - Exteroception: Force sensors and machine vision
- A symbiotic relation
  - Close interaction between the human and robot
- Compact, durable and lightweight power source to actuate the whole exoskeleton
  - Still a major challenge for autonomous robotics
- Complex system design and control
  - Varies with application
- Active Control
  - The key requirement is the ability to adapt to uncertainties
  - In contrast to traditional pure servo type pre-programmed control mechanisms

Main Objectives

Improved design of a human-robot interface

- Sense the bio-signal from human skin surface e.g. EMG
- Apply signal processing algorithms
  - Pattern Recognition
  - Estimation
- Enable the controller to initiate the appropriate movement
- To provide mechanical assistance to human body

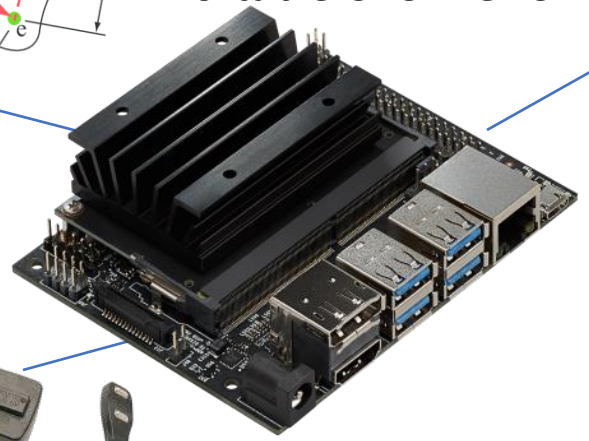
Intelligent Control for EXO-LIMBS



Actuator



Portable CPU + GPU



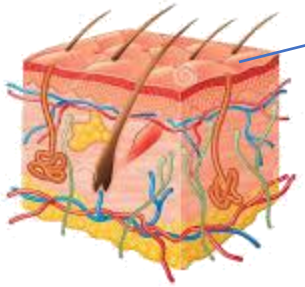
sEMG sensors



DL , SSP &
Control Algorithms



Exo Arm

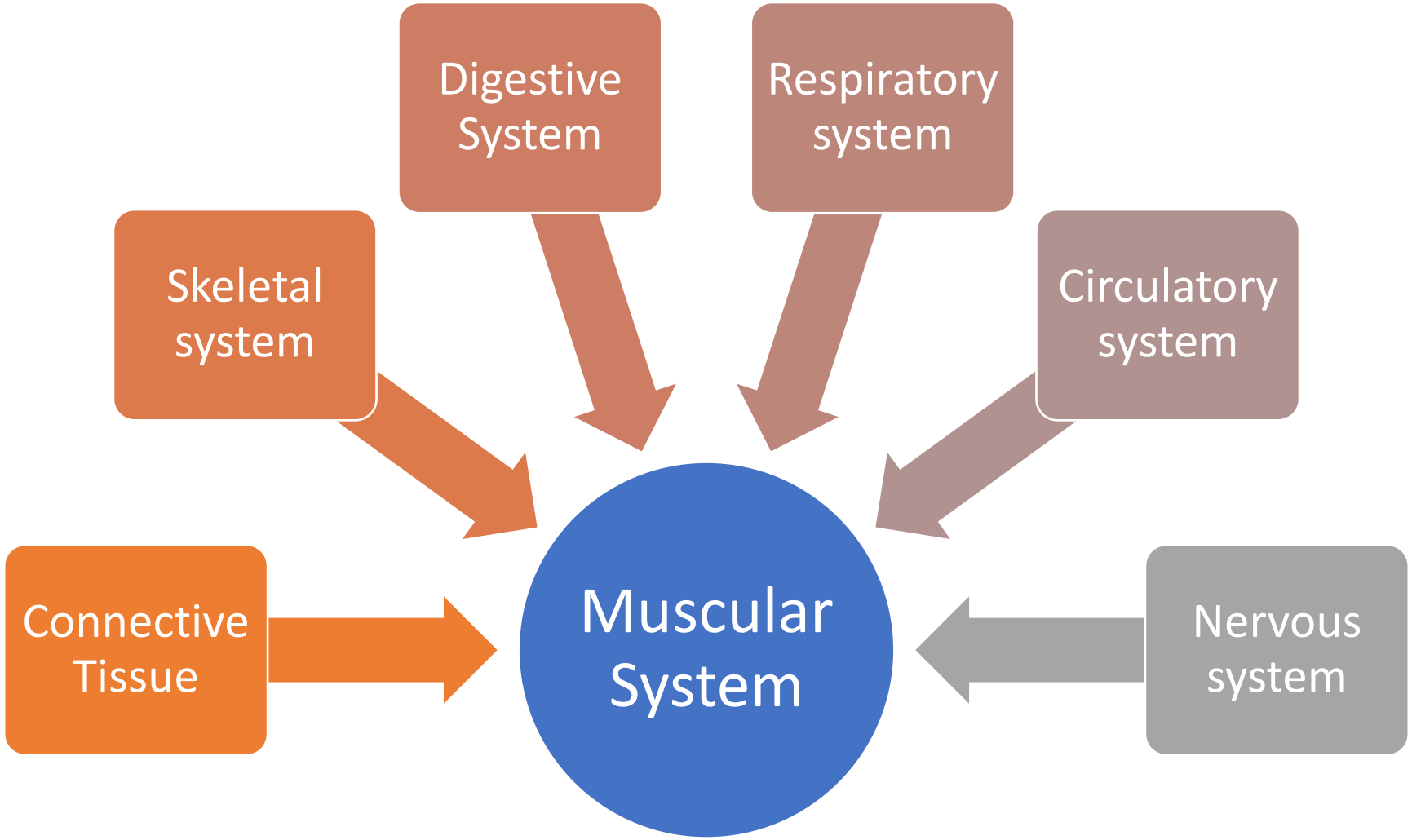


Skin Surface

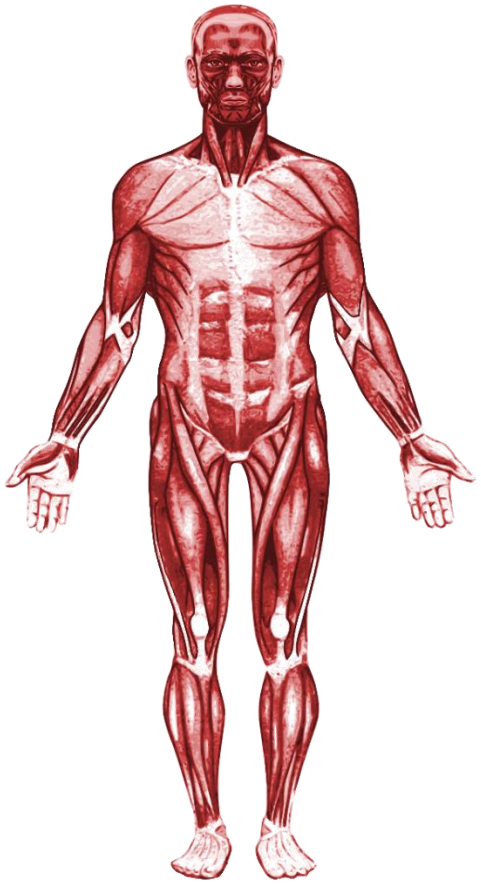
- Bio-signal processing for enhancing the functionality of the exo-skeleton

ANATOMY & PHYSIOLOGY

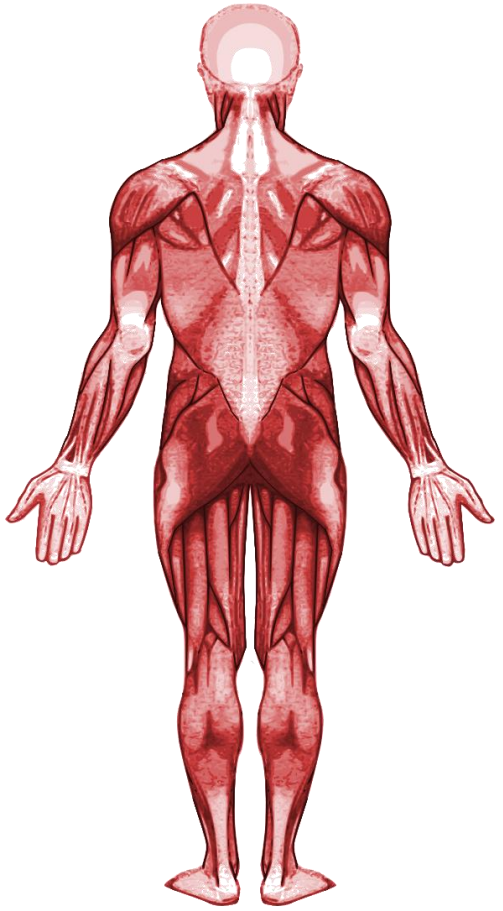
Overview of Neuromuscular system



Human Muscular System



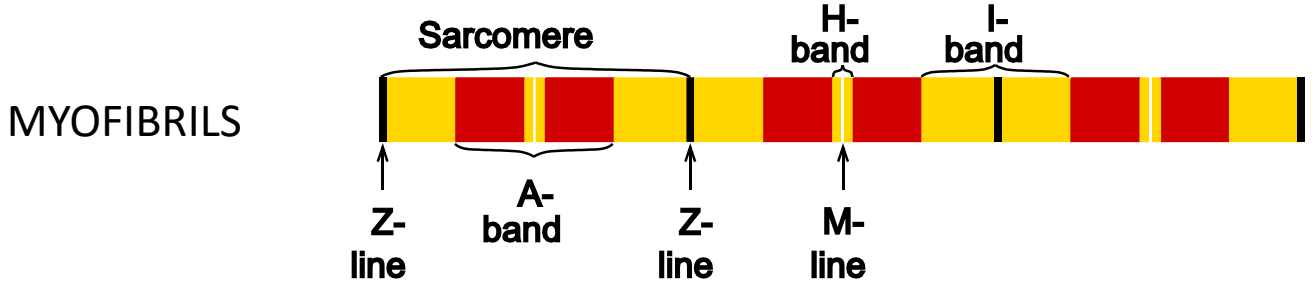
- Normally, Skeletal Muscles make 40% of body weight
- They are the major consumers of energy and metabolism
- They provide structure, maintain posture, cause movements and express emotions



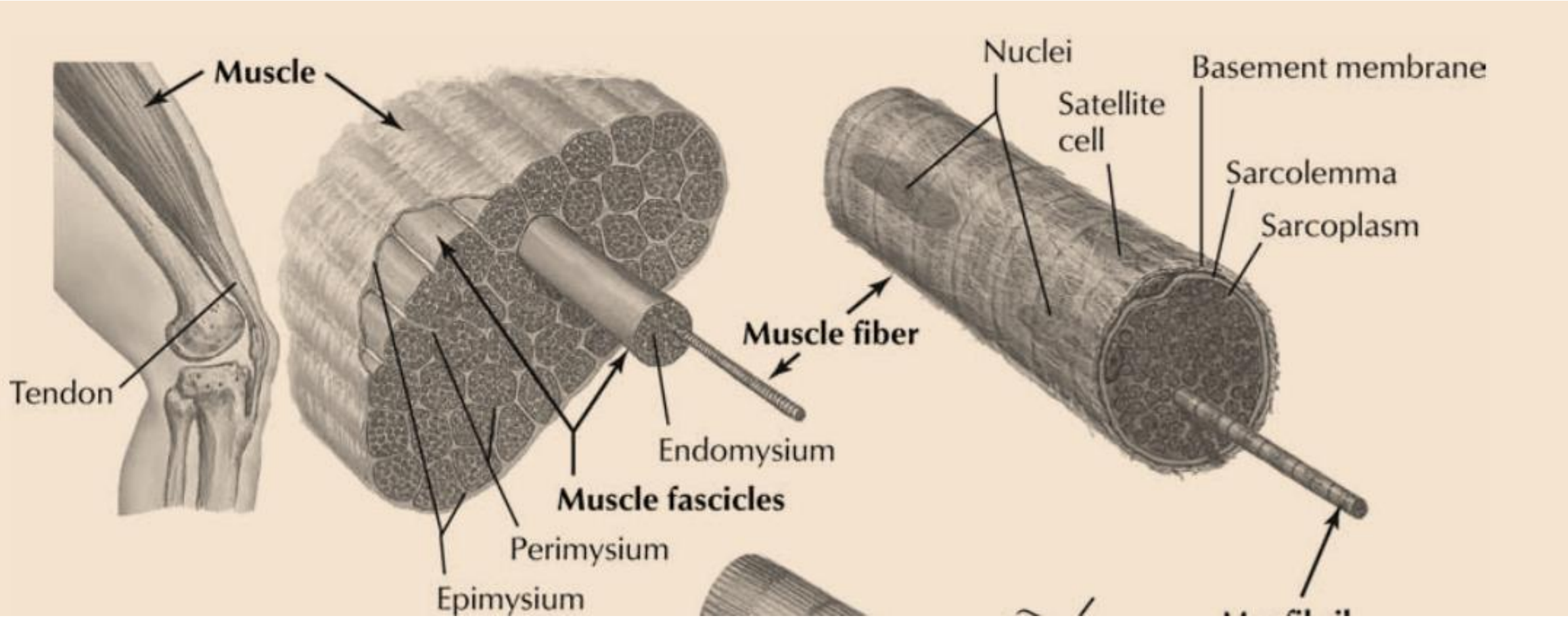
Source: Human Anatomy, Kent M. Van De Graaff , 6th Edition, McGraw Hill Pubs, 2002

Overview of Muscle Function

- Muscle cells are known as myofibrils
- They provide actuation by shortening their resting length (contraction)
- Cause: Ratcheting of myosin fiber against the actin fiber
- Adenosine Triphosphate (ATP) is the fuel for the muscle cells
- Oxygen and Glucose are required to build ATP
- After contraction lactic acid forms and needs to be removed
- Muscle fatigue sets in if this process is not efficient

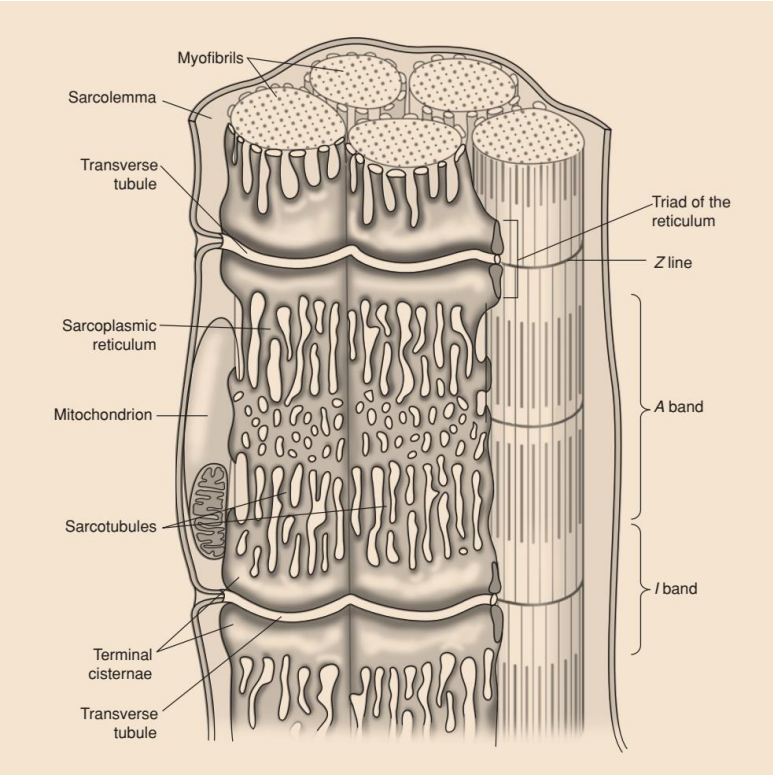


Muscle Anatomical Hierarchy



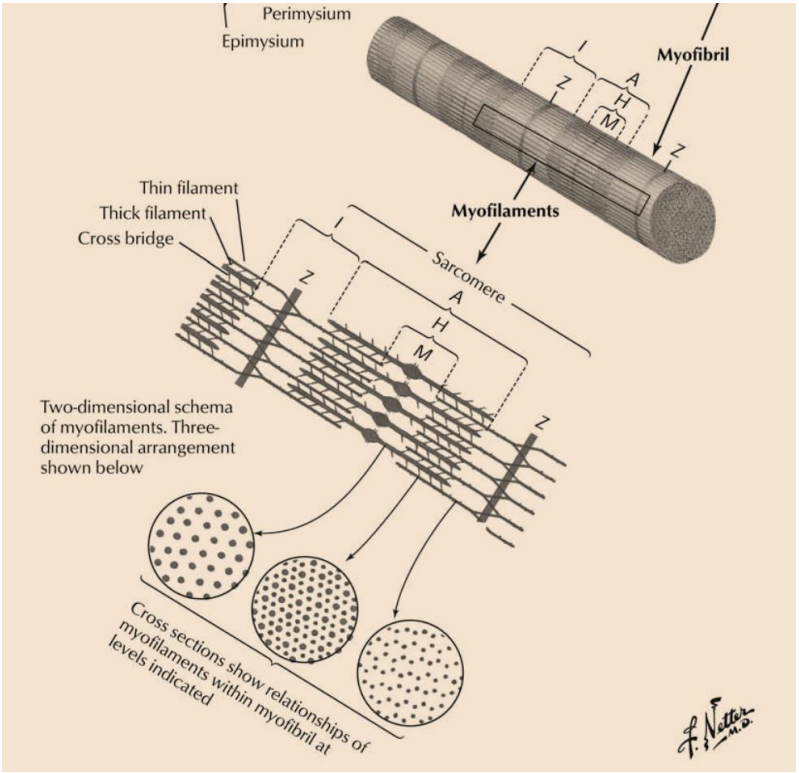
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Microstructure of Myofibrils



The system of transverse tubules

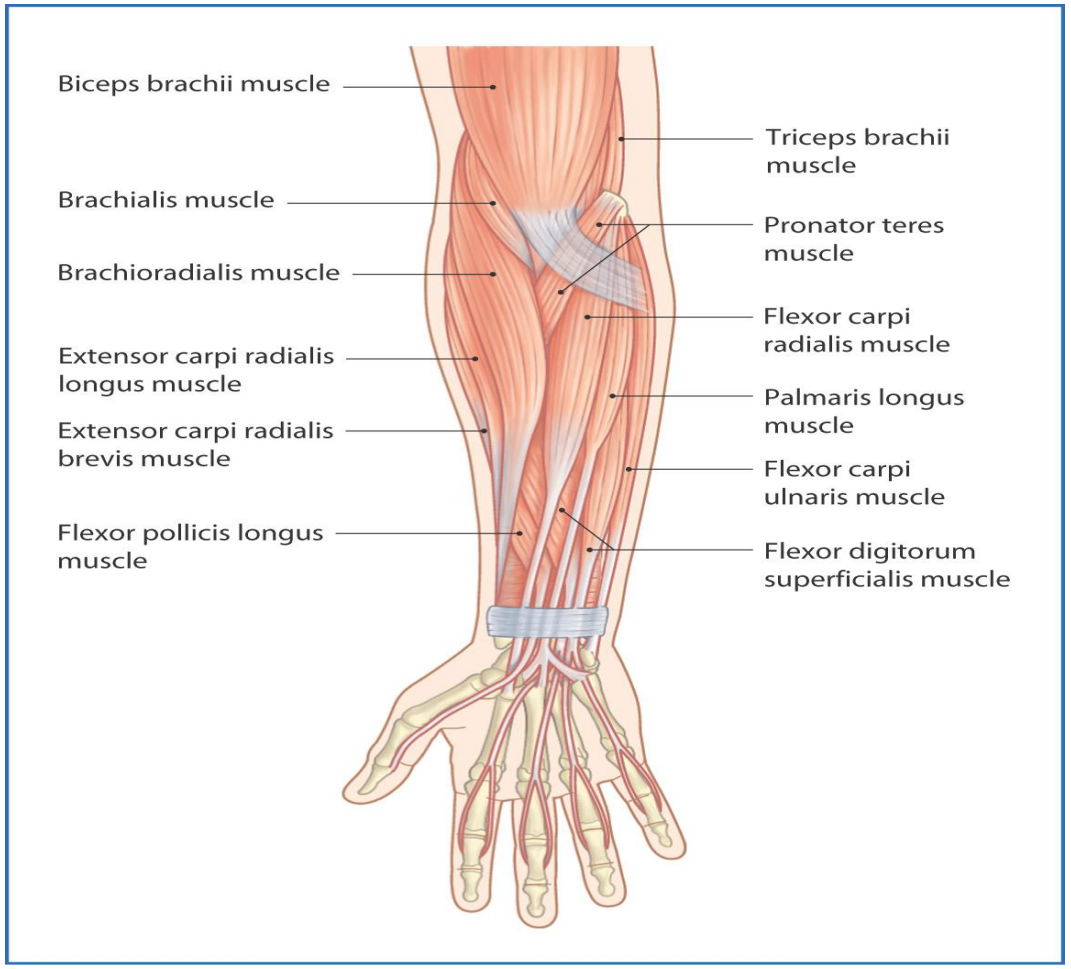
Source: Adapted from W. Bloom and D.W. Fawcett, *A Textbook of Histology*, © 1970, Chapman & Hall.



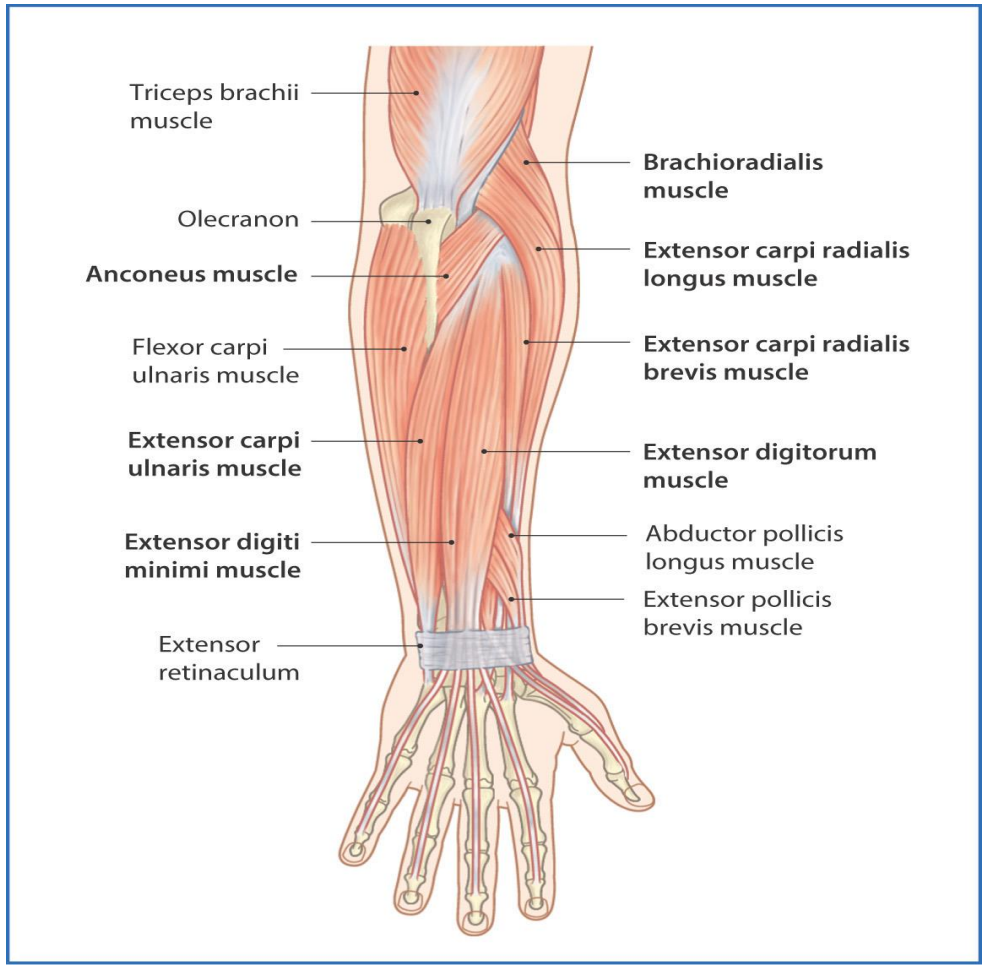
Micro structure of myofibrils

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Anterior View

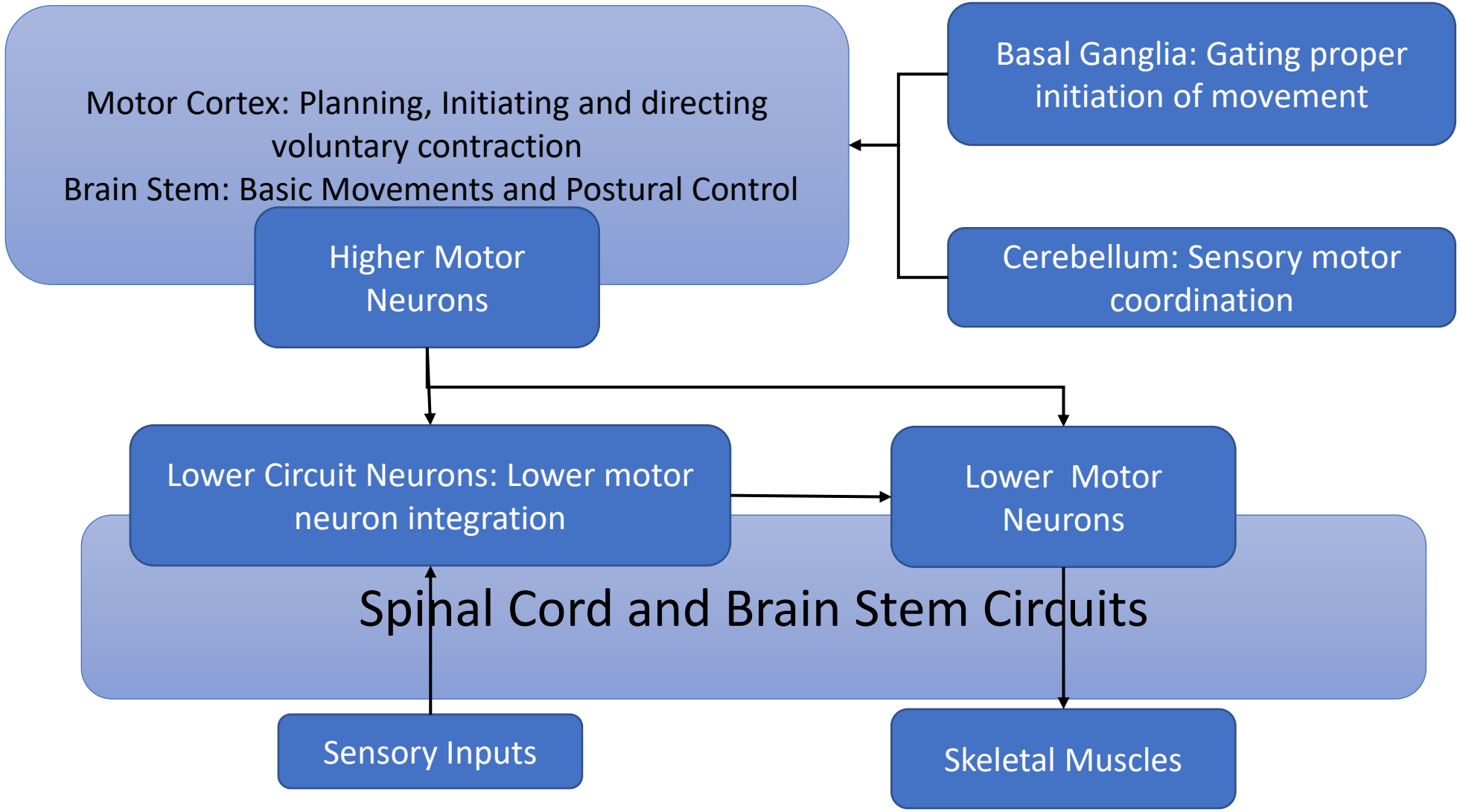


Posterior View

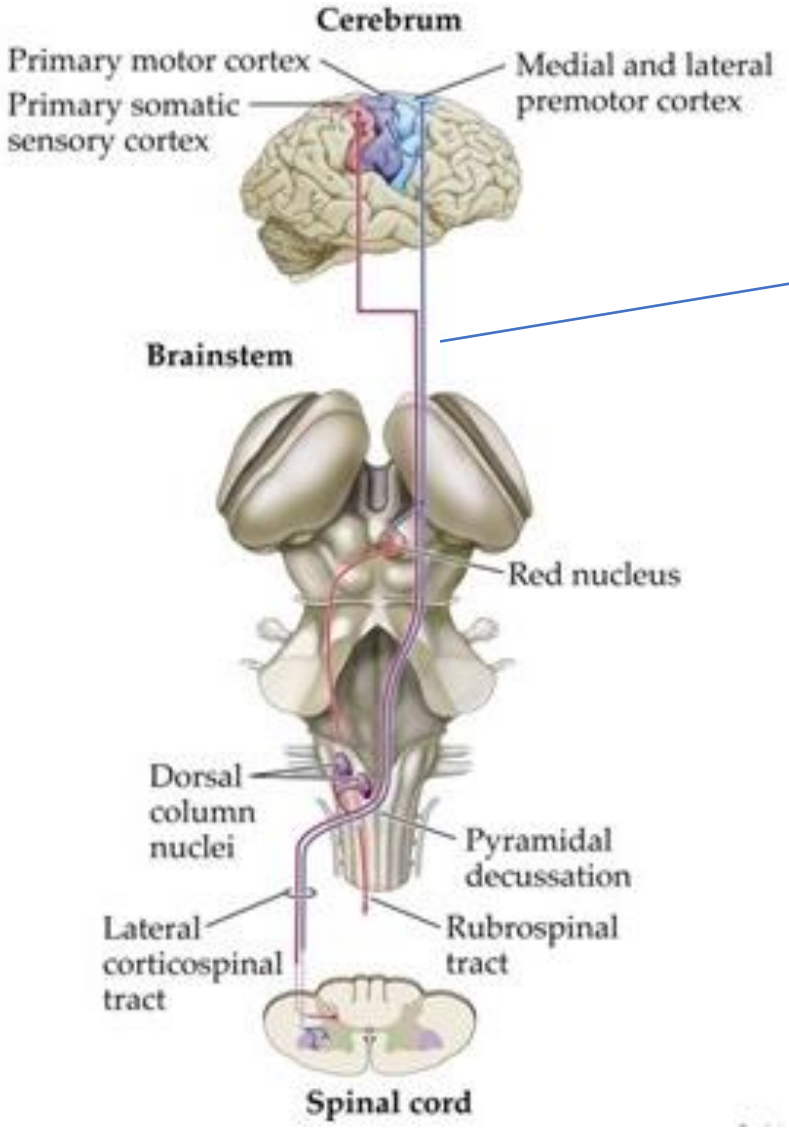


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Motor Control – Signal Flow

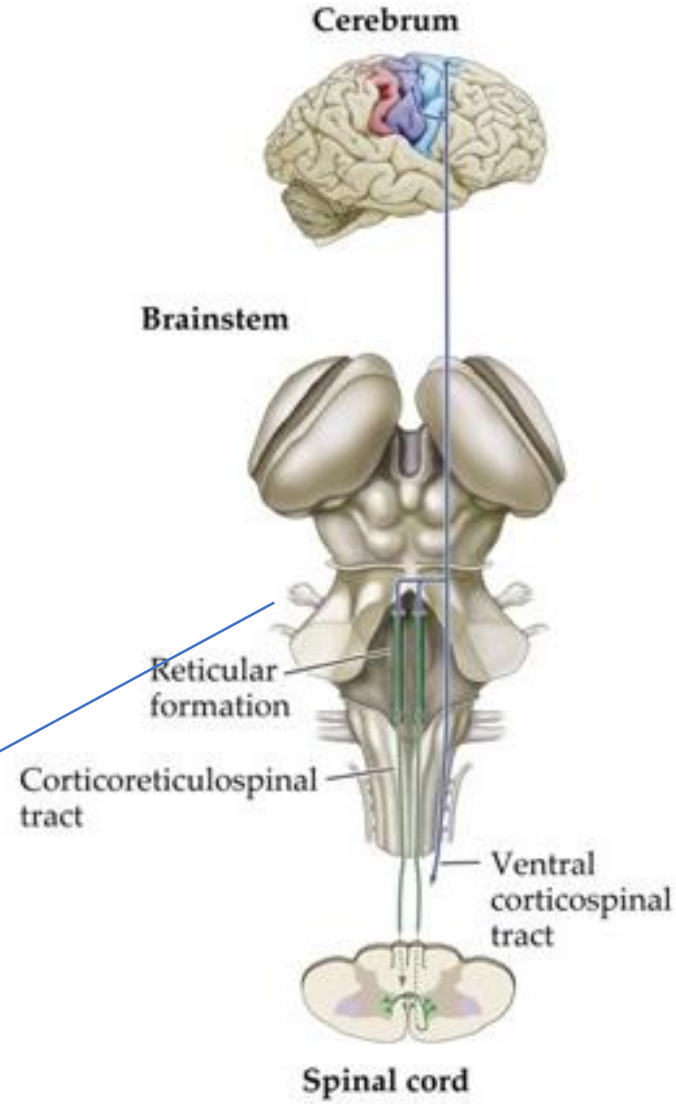


Motor Pathways



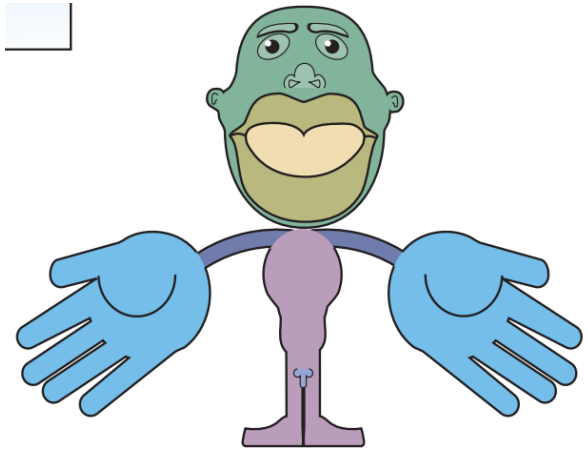
Motor Pathways

1. Direct path from motor cortex, ventral lateral spinal cord: To control hand movements
2. Indirect path: motor cortex through reticular formation and finally to medial spinal cord: to control posture

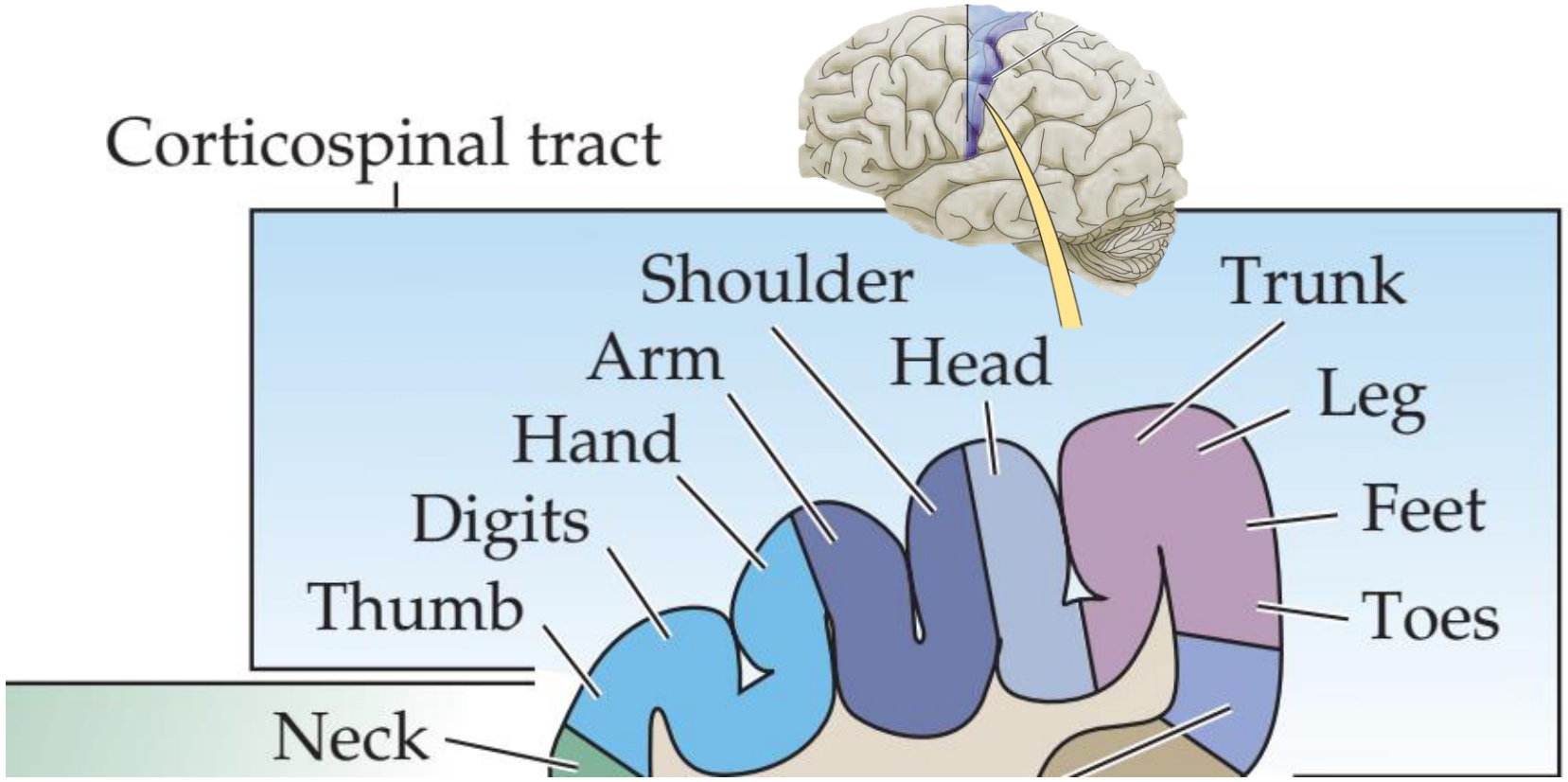


PMC: Sub-Division of Muscle Control

Motor Map: The PMC has a map of the human body's musculature

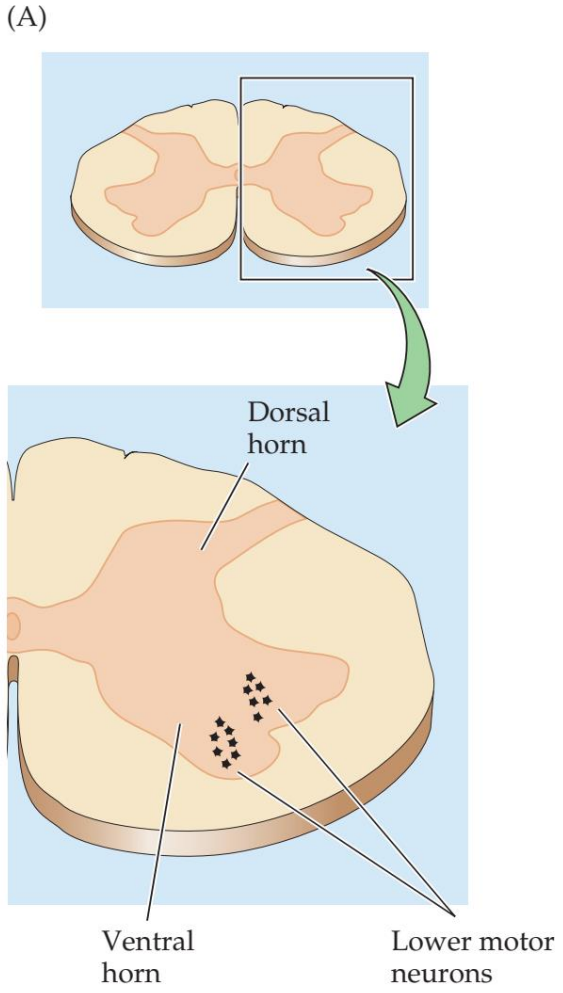


Disproportionate representation of muscle groups in PMC

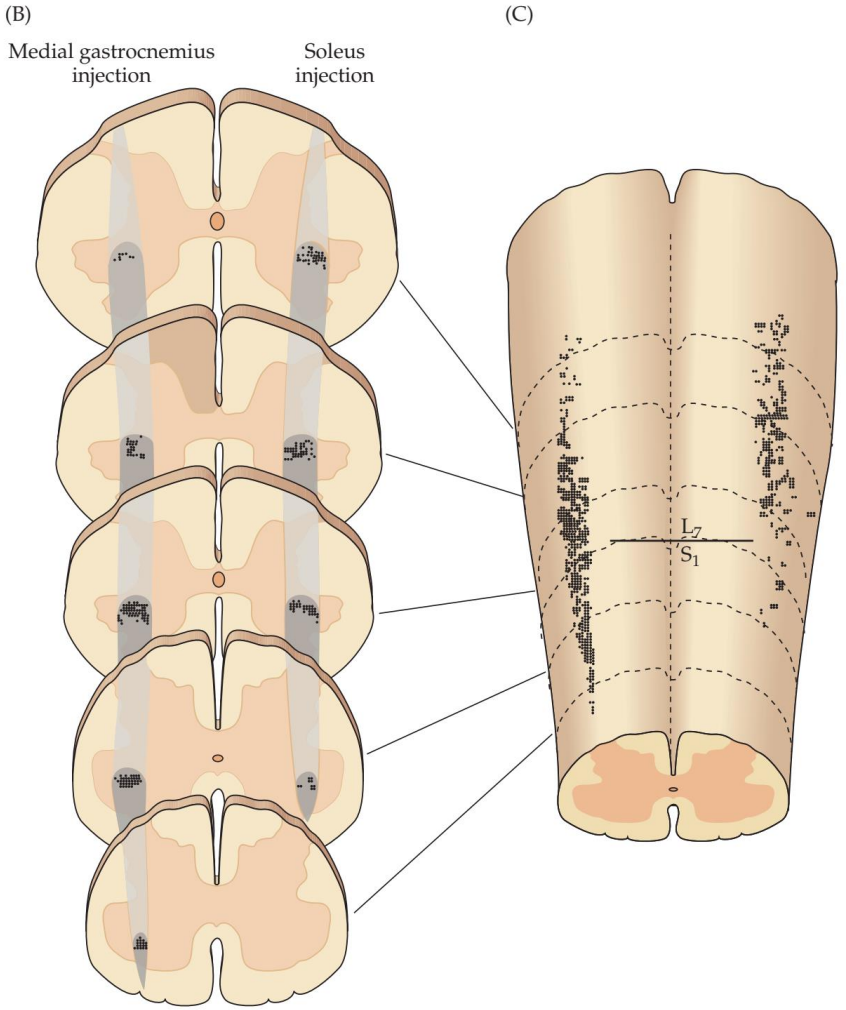


Section along the precentral gyrus, illustrating the somatotopic organization of the motor cortex.

Lower Motor Neurons



ventral horn of the spinal cord



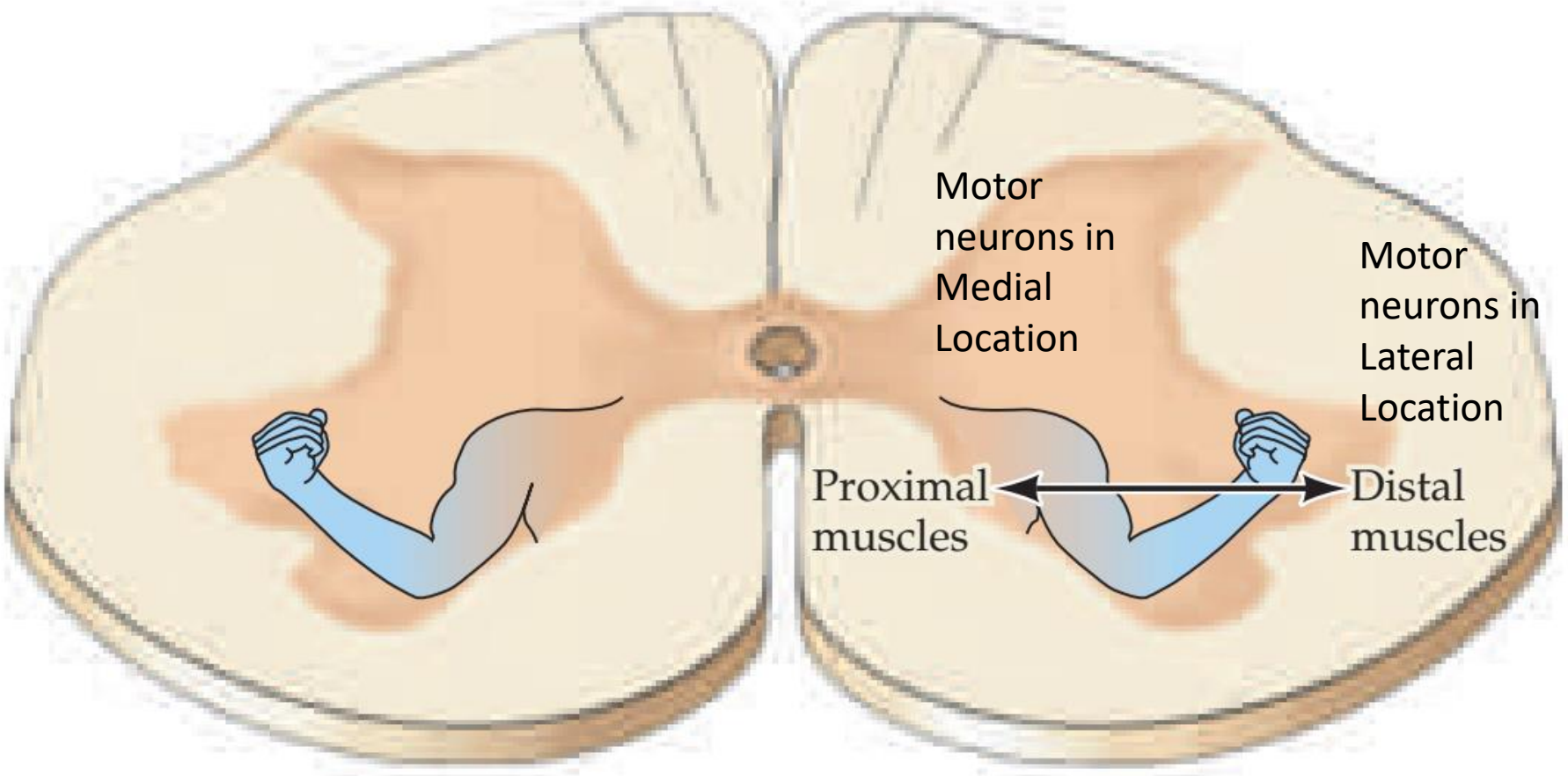
the dorsal view of the reconstructed cord

Structural Information

- A motor neuron (α neuron) innervates fibers in a single muscle
- Several neurons innervating a muscle form a motor neuron pool
- These pools are grouped into rod shaped clusters running parallel to the spinal cord

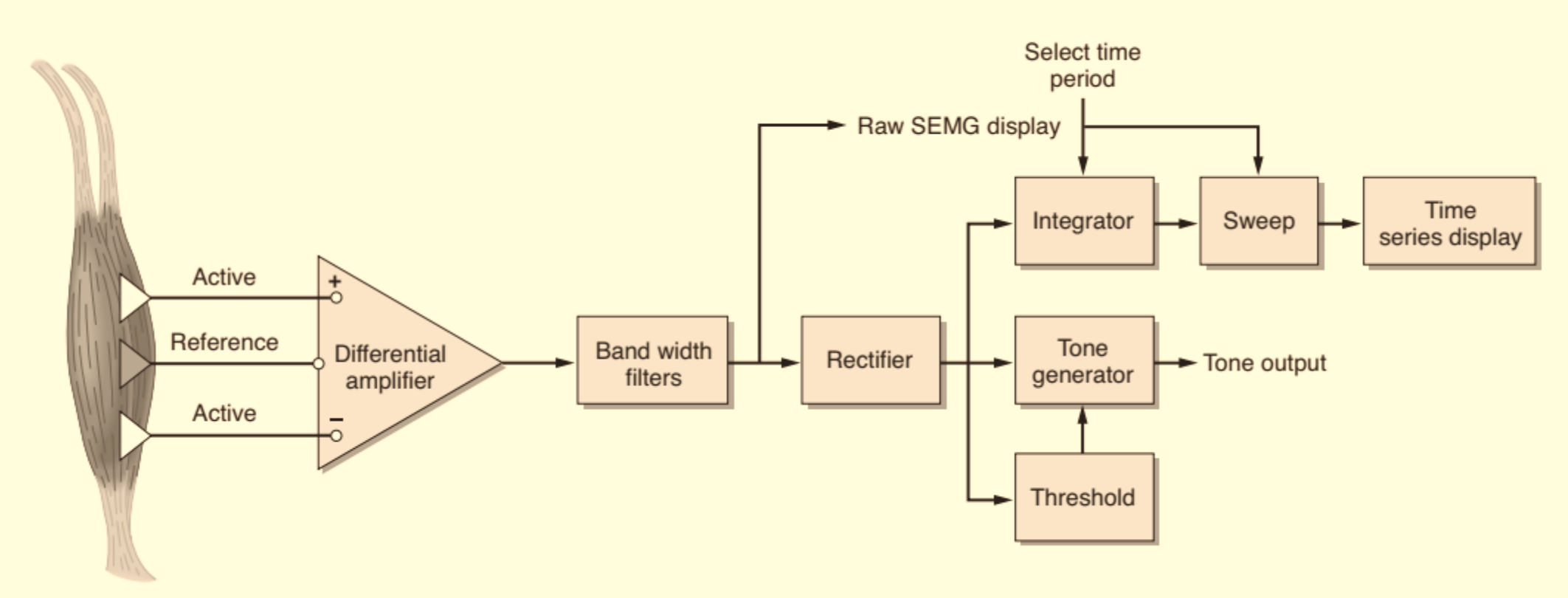
Motor Neuron Mapping to Muscles

Cross section of the ventral horn at the cervical level of the spinal cord



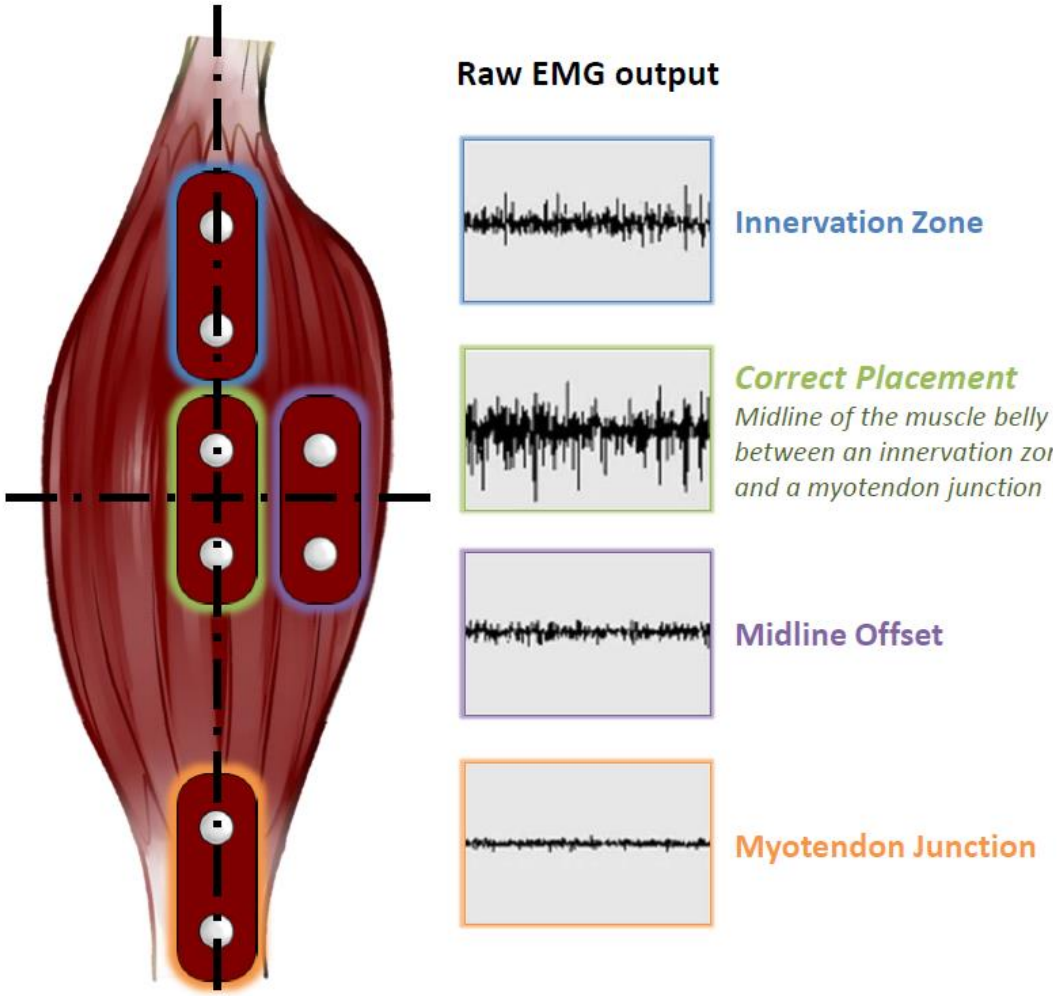
FUNDAMENTALS OF SURFACE EMG SIGNALS

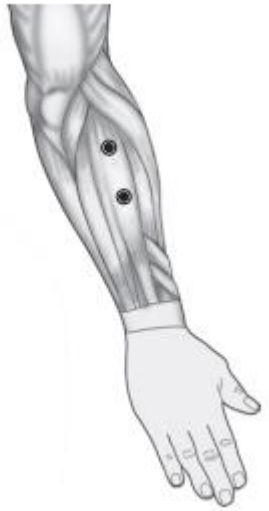
sEMG signal Acquisition



Source: Cram's Introduction to surface Electromyography. 2nd Ed., E. Criswell, Jones and Bartlett Publishers

Electrode Placement





Forearm Extensor Site



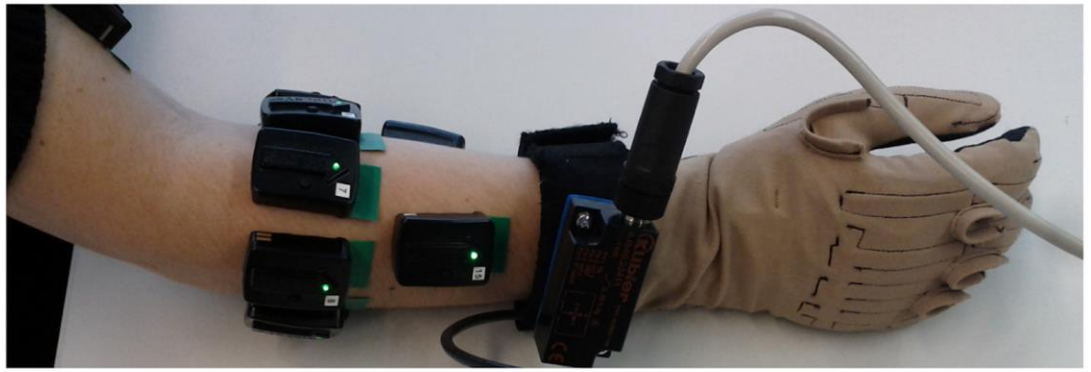
Forearm Flexor Site

Source: Copyright © Clinical Resources Inc.
Ref: Cram's Introduction to surface Electromyography. 2nd Ed., E. Criswell, Jones and Bartlett Publishers
Chapter 17: Electrode Placements

NinaPro Dataset Aquisition [15]



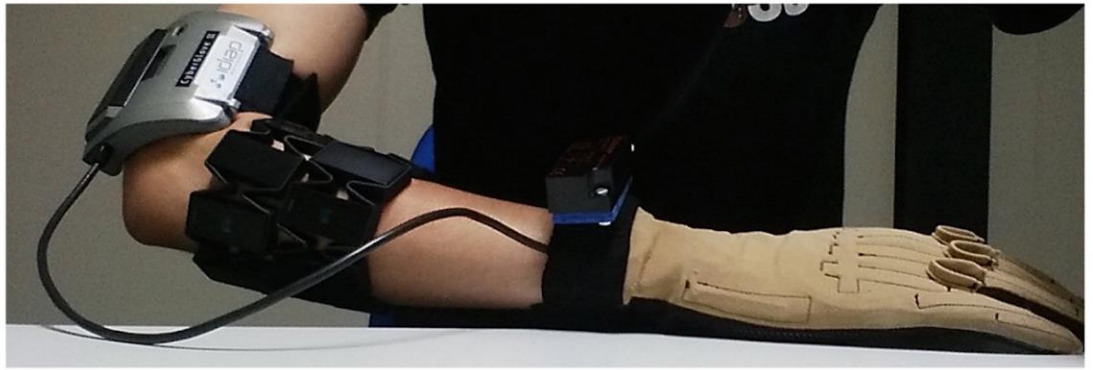
(a) Otto Bock 13 E200 setup



(b) Delsys Trigno setup



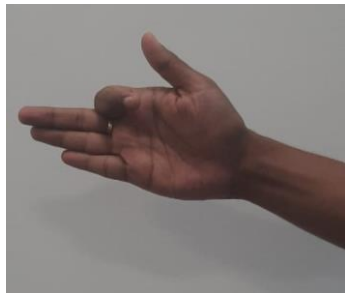
(c) Cometa + Dormo setup

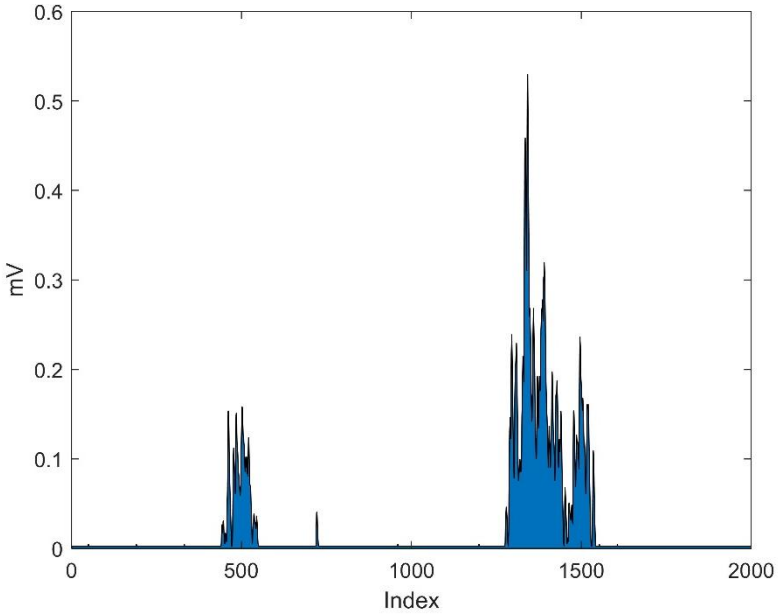


(d) Double Myo setup

Sample of EMG signals

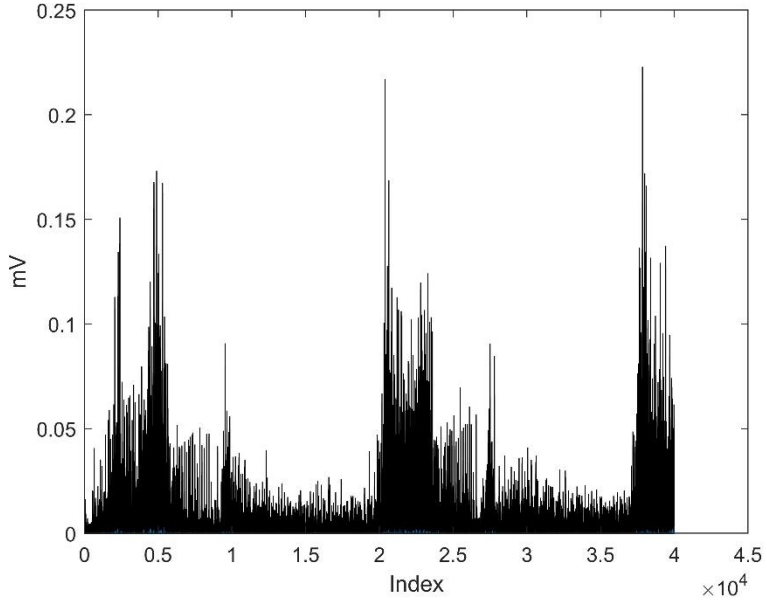
Index Flexion





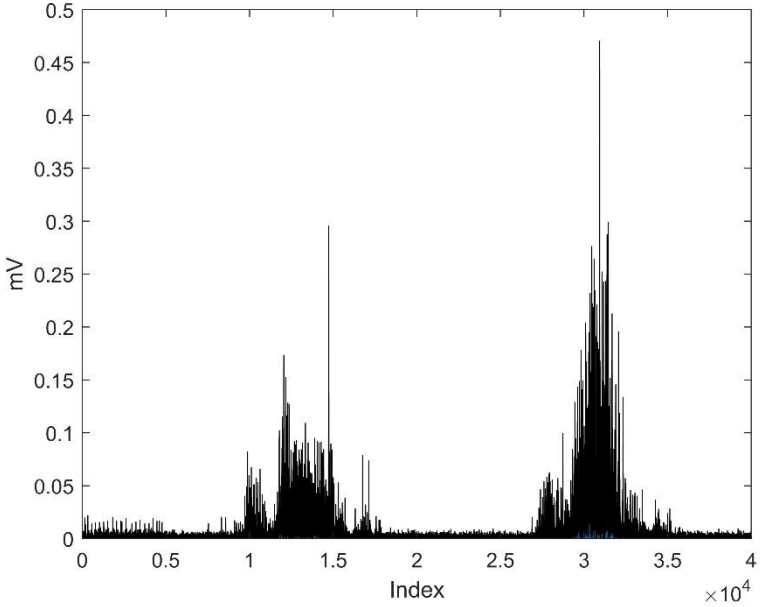
Wrist Extension





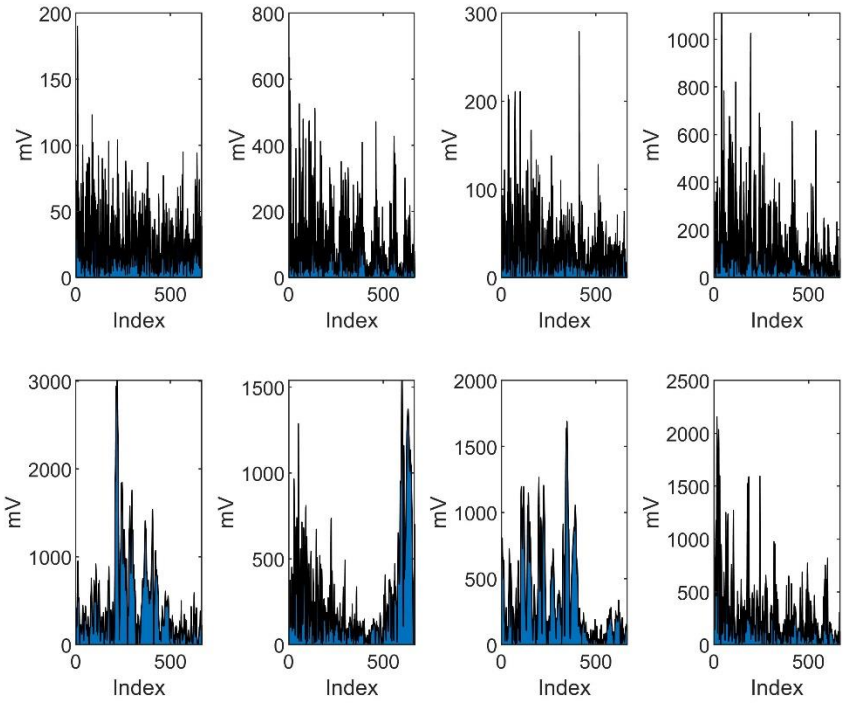
Grasping



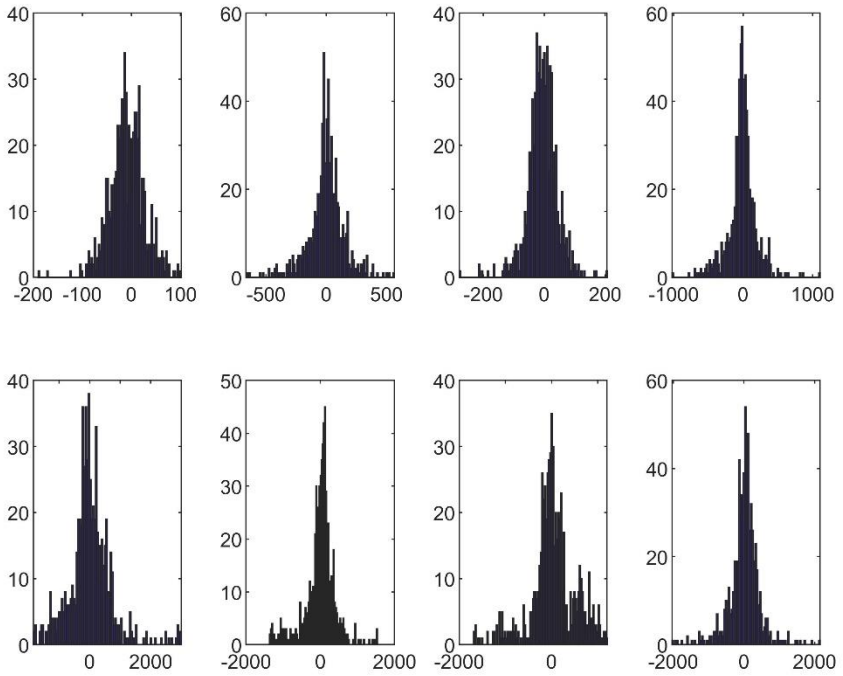


Multi-channel EMG signal

Time series plots of EMG signal



Histograms of EMG signals



Activity: Bowling. Source: Physical Actions Dataset

SURFACE EMG DATASETS

Ninapro Dataset (DB1, DB4) Description

- The ninapro dataset consists of 52 classes of hand gestures [10, 15].
- The dataset is collected from 27 and 10 subjects for DB1 and DB4 respectively.
- The data is collected from 10 Otto bock sEMG sensors placed on hand for DB1 and 12 Cometa miniwave sEMG sensors for DB4.
- The data is sampled at 100Hz and 2 KHz frequency for DB1 and DB4 respectively.
- Each activity is done for 5s of duration with 3s of rest and repeated for 10 times and 6 times for DB1 and DB4 respectively.

Label	Gesture
0	Rest
1-12	Individual finger extension or flexion
13-20	Isometric and isotonic configurations
20-29	Wrist movements
30-52	Grasps and functional movements

Ninapro Dataset (DB1, DB4) Description

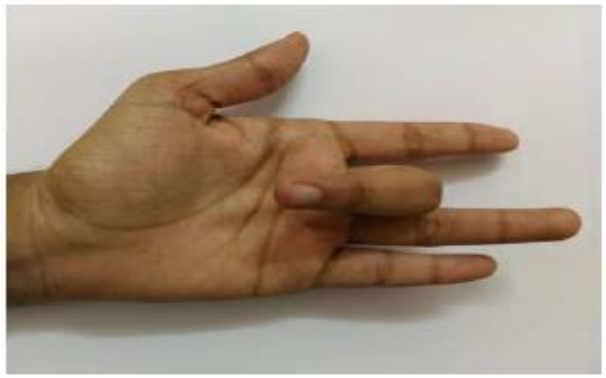
Exercise A:- 12 classes of single finger extension and flexion movements [10].



1. Thumb Flexion



2. Index Flexion



3. Middle Flexion



4. Ring Flexion

Ninapro Dataset (DB1, DB4) Description (Contd…)

Exercise B:- 8 classes of isometric and isotonic finger movements [10].



1)Thumb up
Index finger



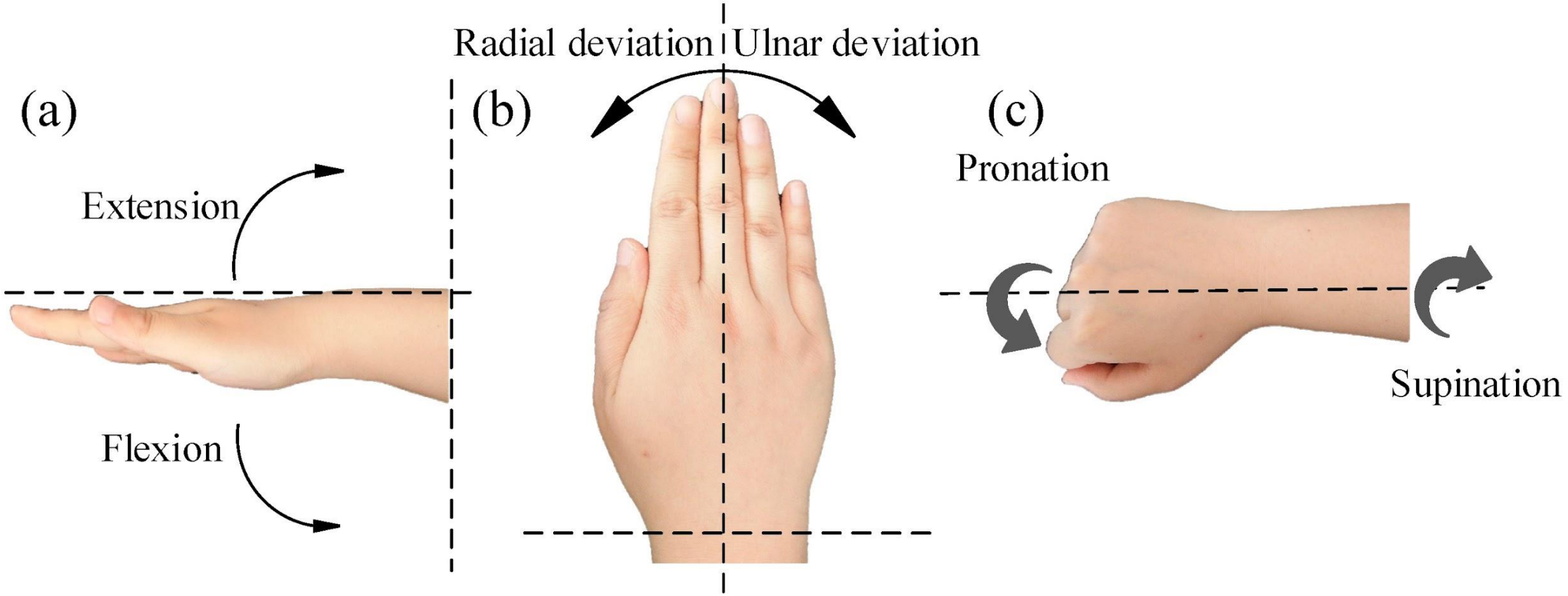
2) Three finger extension



3) Pointing

Ninapro Dataset (DB1, DB4) Description (Contd…)

Exercise B:- 9 classes of wrist movements [17].



Ninapro Dataset (DB1, DB4) Description (Contd…)

Exercise C:- 23 classes of hand grasp functions [10].

	Prismatic	Round	Flat
Power	 Large Wrap  Small Wrap  Index Finger Extension	 Power Sphere	 Extension Type
Precision	 Writing Tripod  Thumb-n Finger	 Precision Sphere	 Lateral Pinch

Characteristics of Datasets [10,15]

Dataset	No. of classes	No. of subjects	No.of Trials per class	Total no. of samples	Sampling Frequency	Sensor name	No. of electrodes/ channels
DB1	52	27	10	14,040	100 Hz	Ottobock	10
DB2	49	40	6	11,760	2 kHz	Delsys Trigno Wireless	12
DB4	52	10	6	3120	2 kHz	Cometa Miniwave sensors	12

Characteristics of Datasets [18,19]

Dataset	No. of classes	No. of subjects	No.of Trials per class	Total no. of samples	Sampling Frequency	Sensor name	No. of electrodes/ channels
DB 5	52	10	6	3120	2 KHz	Cometa Miniwave sensors	12
CapgMyo DBa	8	18	180	1440	1 KHz	silver wet electrodes	8 X 16
CSL-HDEMG	27	5	10	1350	2048 Hz	OTTO Bioelectronica	7 X 24

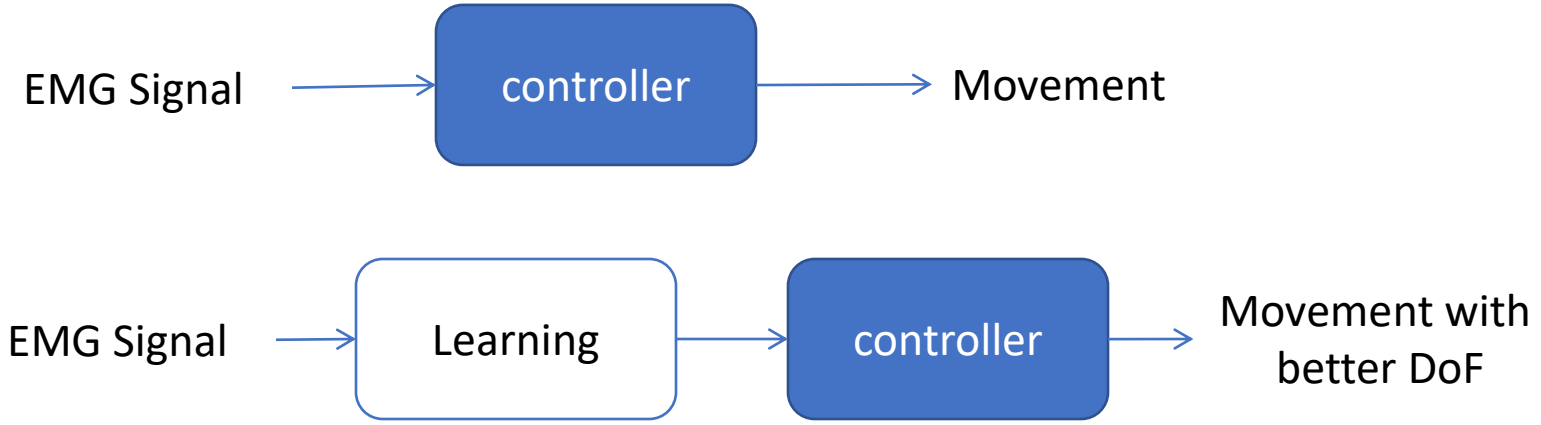
Characteristics of Datasets [18,19]

Dataset	No. of classes	No. of subjects	No.of Trials per class	Total no. of samples	Sampling Frequency	Sensor name	No. of electrodes/ channels
BioPatRec	26	17	10	4420	2 KHz	Thalmic Myo Armband	4
Myo Dataset	7	19	4	532	200 Hz	Myo Armband	8

EMG CLASSIFICATION

Need for Learning-Enabled Controller

- Goal is to learn the intention of the user
- Provides an intuitive/natural control of the robot
- Learning based controllers are available in the market
- Only 10% of US amputees have access



Considerations for Learning Based Controller

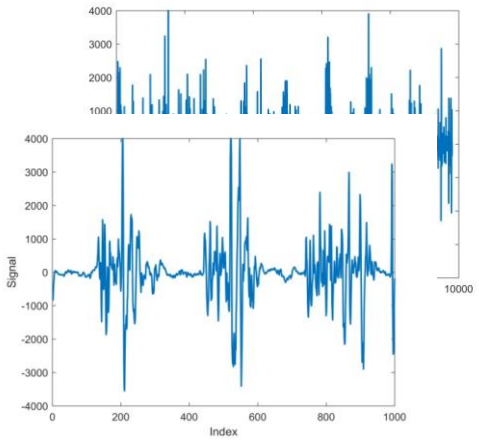
Challenges

- Work in real time control applications
- Provide maximum degrees of freedom in movement
- Lack of physiologically causal musculature
- EMG Crosstalk
- sEMG may not be sufficiently informative
- How much classification accuracy is enough

Solutions

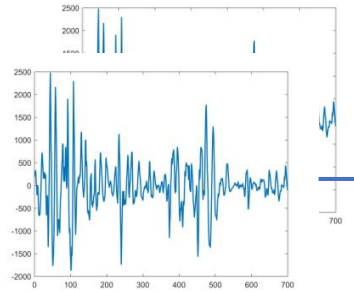
- Multi-channel EMG
- Surgery to provide access to neural pathways
- Test and improve in virtual environment
- Accuracy vs. usability may have a trade-off

sEMG Classification



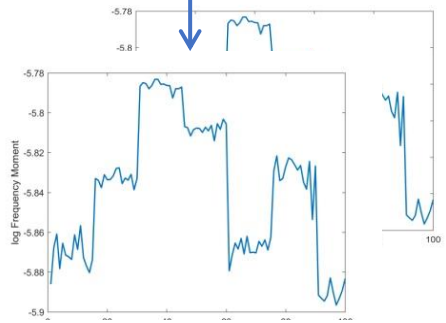
EMG Sequence

SEGMENTATION & PREPROCESSING



EMG Segments

FEATURE EXTRACTION



Frequency Domain features

CLASSIFIER

LIMB MOVEMENT DIRECTION

Left vs. Right
Up vs. Down

Desirable Characteristics

Relevant Features

- Provide unique information for particular classification application
- A robust feature set that provides maximum efficiency

Robustness against

- Electrode shift
- Muscle fatigue
- Muscle contraction effort
- Force variation
- Limb position

CLASSICAL APPROACHES

Features for EMG learning

- Mean absolute Value (MAV)
- Zero crossings
- Slope sign changes
- Waveform length
- Willison amplitude
- Root mean square
- Variance
- Log-detector
- AR Coefficients
- Mean Frequency
- Median Frequency
- Peak Frequency
- Mean power
- Wavelet based features
- Band powers from PSD
- Cepstral features
- Moments from FFT
- Ratio of moments

Physical Actions Dataset







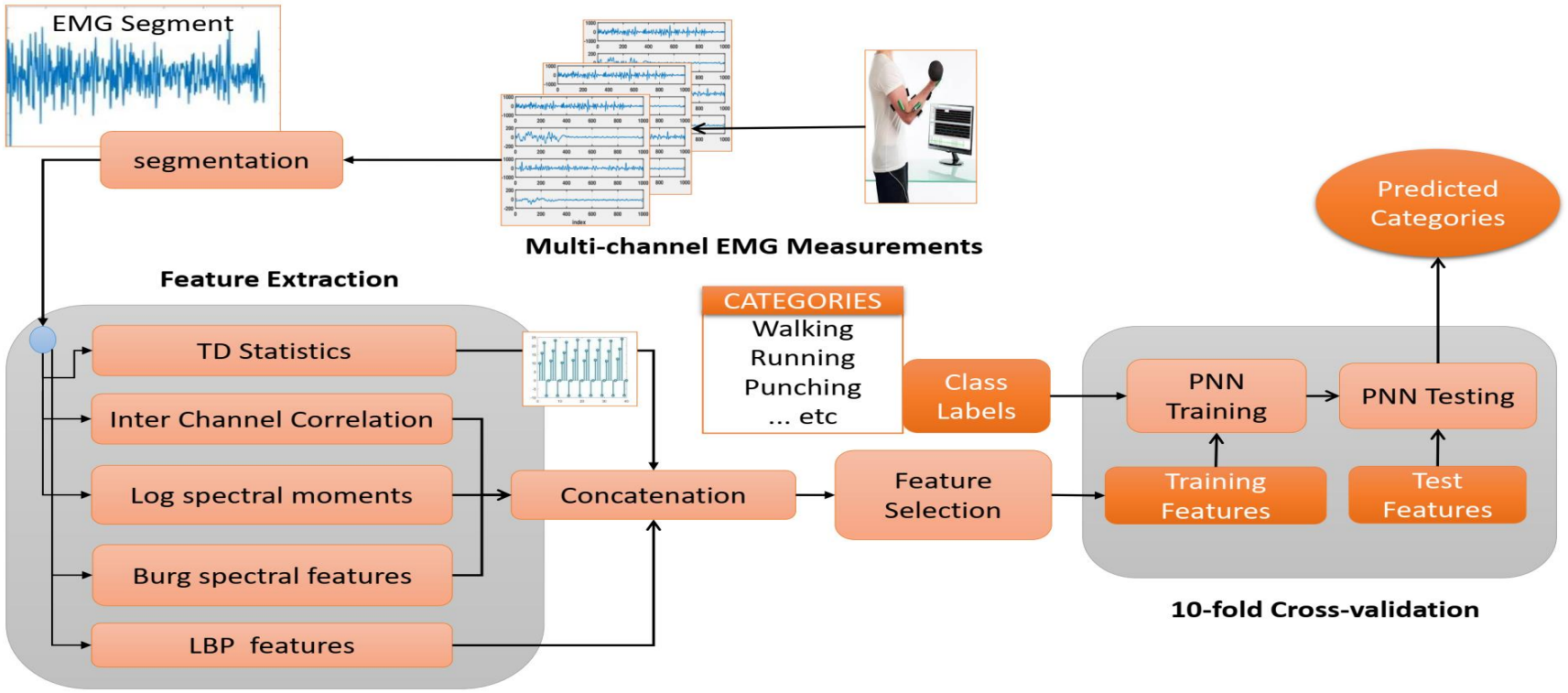




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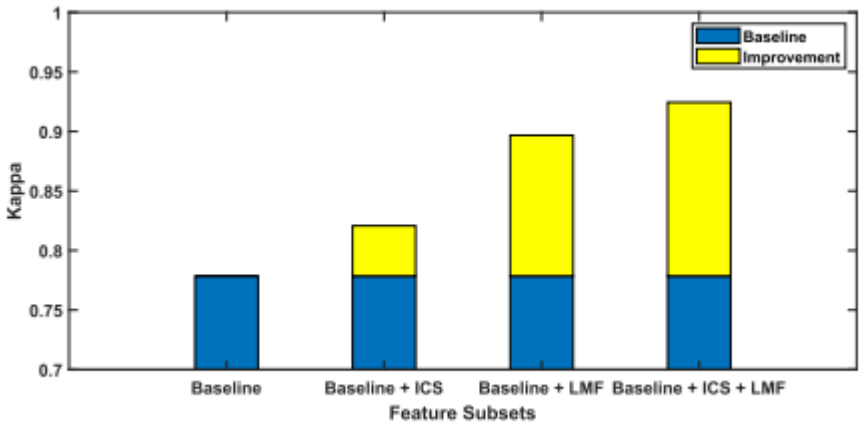
NN with Hand-designed Features

➔ Turlapaty et al. (2019), [2] extracted various EMG features for movement classification.

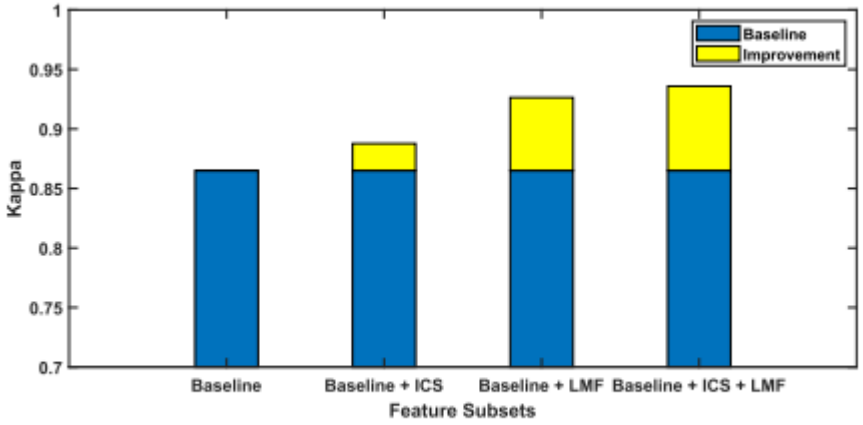


Performance Analysis

Accuracy Improvement

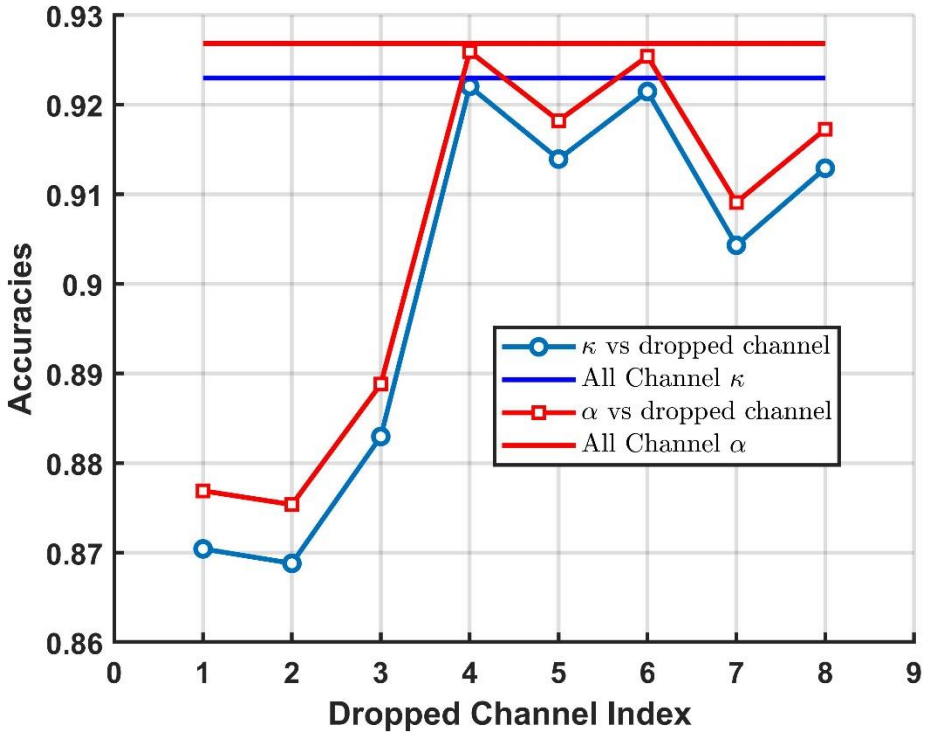


(a) PNN



(b) sKNN

Channel Relevance



Limitations on Existing Tools

- Lack of large scale sEMG dataset.
- No dataset with Indian population.
- Not very much high accuracy achieved.
- No neural network architecture specifically for sEMG signal classification.

DEEP LEARNING APPROACH

Challenges

Automated Feature Extraction

Traditional features

- Time domain stats, AR coefficients, spectral band powers and ratios etc.

Objective: To learn robust features (Automated discovery of important features)

- Under variations of age, gender, health conditions
- Electrode shifting, force variations, limb position variations and signal interference

Learning the Weights

- Training
  - Algorithm: Standard mini-batch stochastic gradient descent
  - Hyperparameter selection through validation performance
  - Activation function selection
  - Step delay learning rate
- Testing
  - Validation of the model on unseen data

Performance Evaluation

- Role of Muscle Force (% MVC)
  - Case 1: The subjects perform the N categories of hand movements
  - Case 2: The subjects repeat the above movements while holding a weight such as a dumb-bell
  - Case 3: The subjects repeat the movements in (1) while carrying a common household object
- Role of subjects
  - Need of training on data from new subjects
- Temporal variations
  - On daily and longer scales

Model Generalization

Applicability of the trained model under different conditions as follows

- Geographic location
- Age group
- Occupation

Amount and extent of re-training depends on

- Sample size of new examples
- Similarity of new data with the old data used in training

Signal Processing of Input

Raw EMG signals	• Reject only outliers
Integrated signals	• Window length vs. SNR
Integrated absolute values	• Rectification, weighted averaging etc
Wavelet coefficients of the EMG signals	• Selection of mother wavelet function and levels of decomposition
Short-time Fourier transform of the EMG signals	• Window length for application of STFT

Recent Models and Results

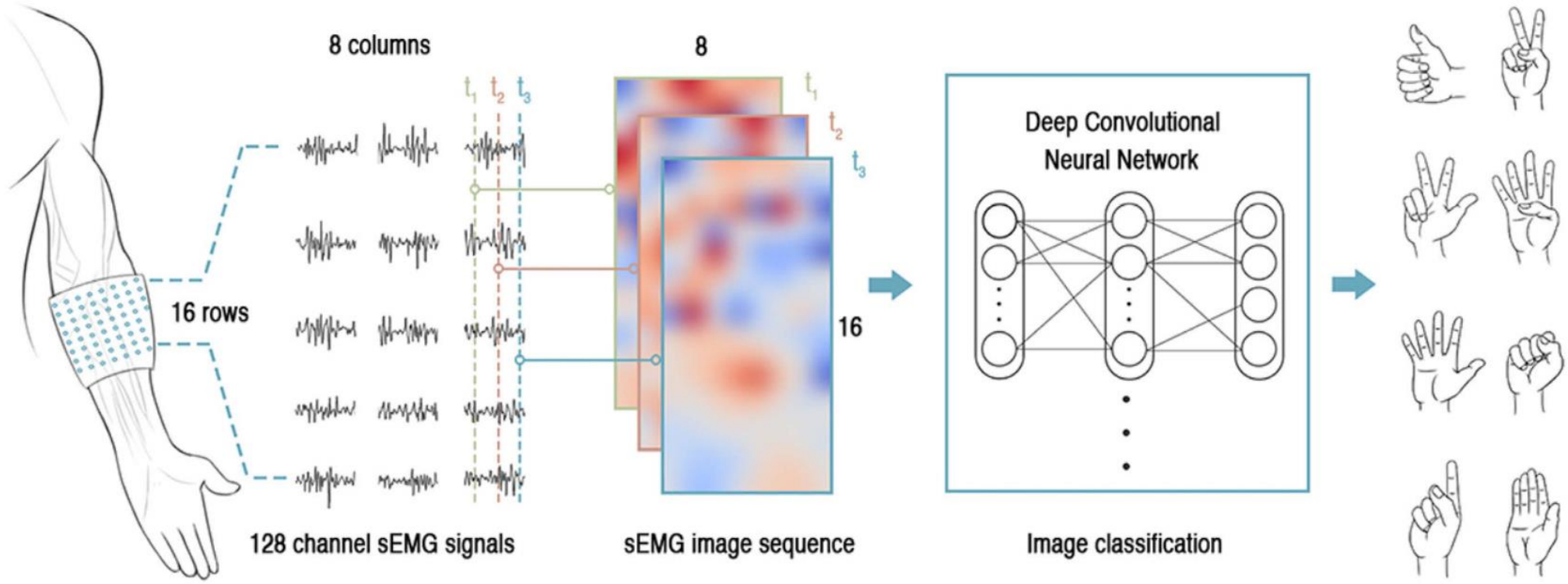
CNNs for sEMG classification

- Geng et al. (2016), [12] has worked on the following datasets
  - NinaPro DB1
  - CSL-HDEMG
  - CapgMyo DB-a

- Considered each sample point as image and performed classification
  - NinapPro DB1 i/p shape is 1 X 10
  - CSL-HDEMG i/p shape is 7 X 24
  - CapgMyo DBa i/p shape is 8 X 16

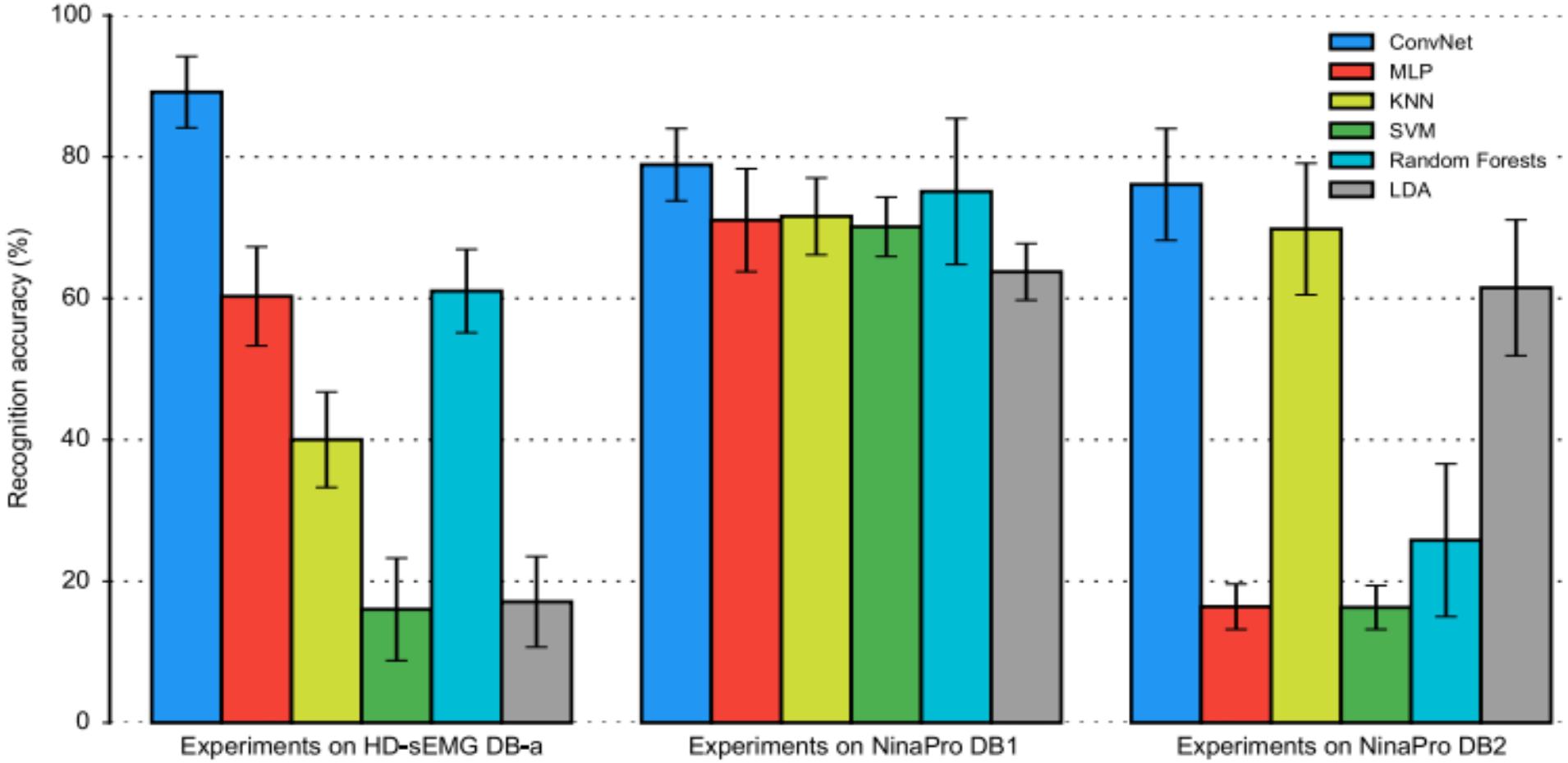
sEMG as an Image

- Geng et al. (2016), [12] has analysed performance of sEMG datasets classification by using instantaneous sEMG images.



CNN Results

➔ 8 hand gestures recognition accuracy.



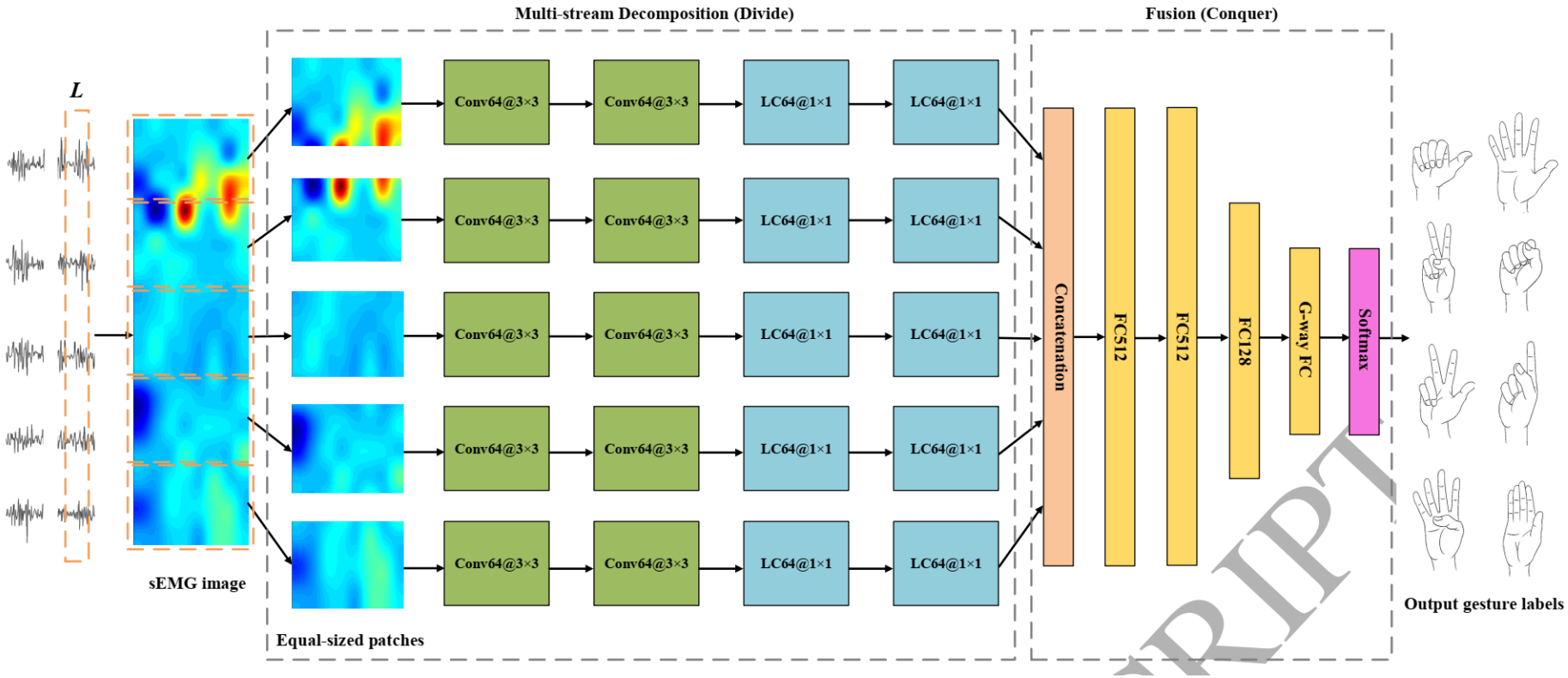
CNN-LC for Gesture classification

➔ Wei et al. (2019), [13] has split each sEMG image into patches for providing to the CNN for further classification.

NinaPro DB1	No. of Streams	CSL-HDEMG	No. of Streams	CapgMyo DB-a	No. of Streams
Channel-wise Ms	10	7 X 24-SS	1	8 X 16-SS	1
hybrid-Ms	3	7 X 1-patch wise MS	24	8 X 2 array wise Ms	8
20X10-SS	1	7 X 2-patch wise MS	12	8 X 4 array wise Ms	4
20X10X1-SS	1	7 X 4-patch wise MS	6	8 X 8 array wise Ms	2
frame-wise-MS	20	7 X 8-patch wise MS	3	1 X 16 patch wise Ms	8
adjacent-frame-MS	19	1 X 24-patch wise MS	7	2 X 16 patch wise Ms	7
		2 X 24-patch wise MS	6		

2CNN-2 LC Architecture

➔ Wei et al. (2019), [13] has split each sEMG image into patches for providing to the CNN for further classification.



2CNN-2LC Results

➔ Wei et al. (2019), [13] Results

NinaPro DB1	Classification Accuracy (%)	CSL-HDEMG	Classification Accuracy (%)	CapgMyo DB-a	Classification Accuracy (%)
Channel-wise Ms	85	7 X 24-SS	89.5	8 X 16-SS	89.2
hybrid-Ms	81.6	7 X 1-patch wise MS	89.5	8 X 2 array wise Ms	89.5
20X10-SS	82.9	7 X 2-patch wise MS	89.6	8 X 4 array wise Ms	89.4
20X10X1-SS	83.3	7 X 4-patch wise MS	89.8	8 X 8 array wise Ms	89.4
frame-wise-MS	79.4	7 X 8-patch wise MS	90.3	1 X 16 patch wise Ms	84.2
adjacent-frame-MS	79.2	1 X 24-patch wise MS	90	2 X 16 patch wise Ms	87.9
		2 X 24-patch wise MS	90.2		

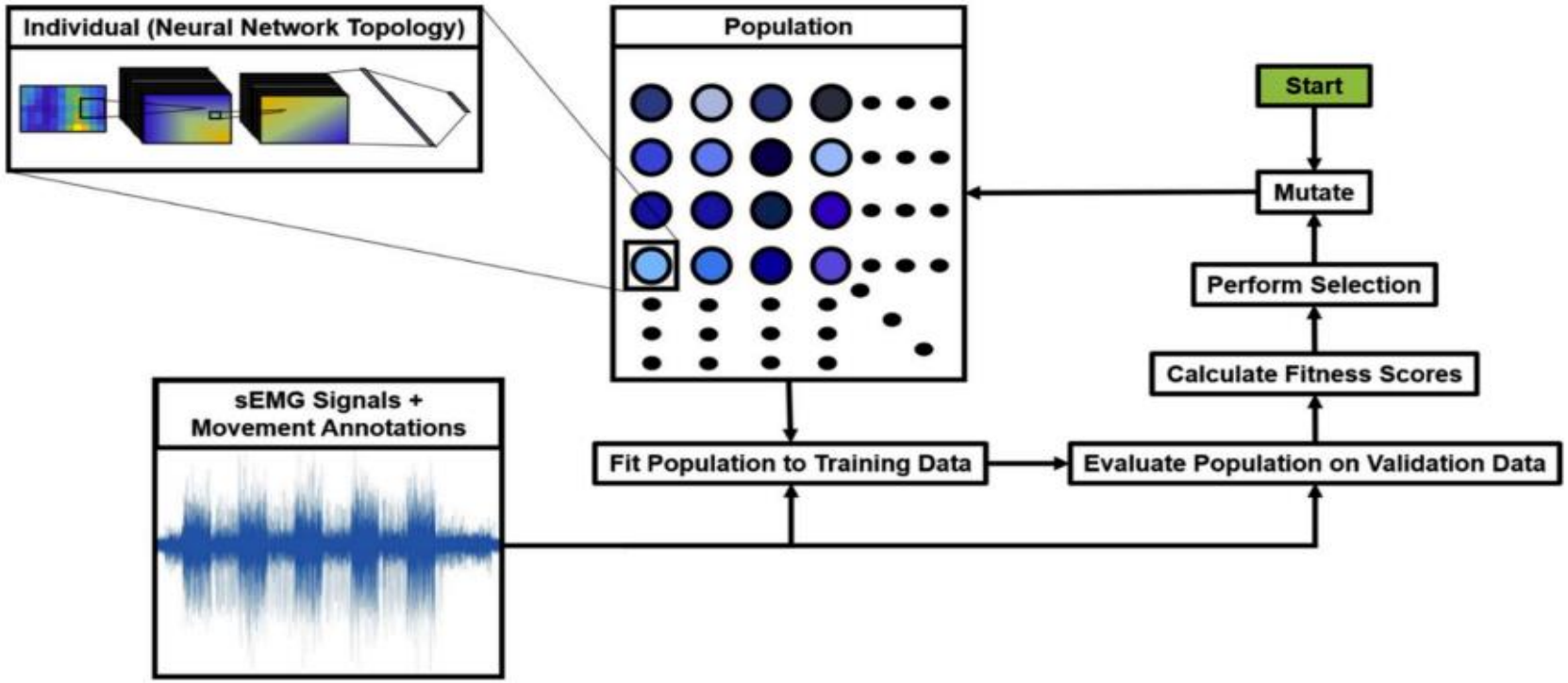
Evolved CNN

- Olsson et al. (2020), [8] has worked on the following datasets
  - NinaPro DB1 and DB2
  - BioPatRec
  - CapgMyo DB-a

- Considered each sample point as image and performed classification
  - NinapPro DB1 and DB2 i/p shape is 100 X 12 X 1
  - BioPatRec i/p shape is 7 X 24
  - CapgMyo DB-a i/p shape is 100 X 8 X 16

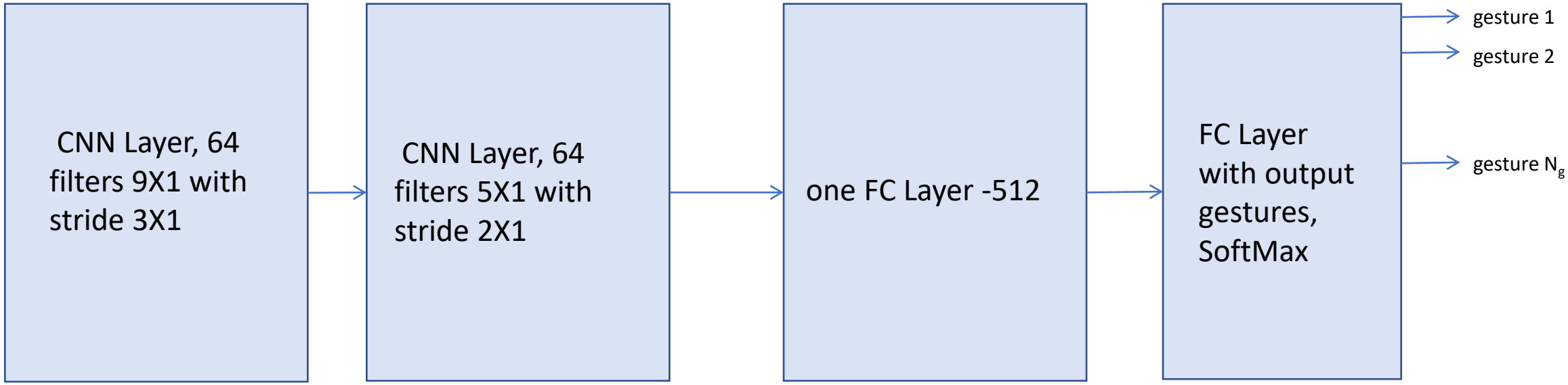
Evolved CNN– NET Topology

➔ Olsson et al. (2020), [8] has proposed an algorithm for identifying the network topology for sEMG classification.



Evolved CNN Architecture

➔ Olsson et al. (2020), [8] CNN architecture



Evolved CNN– Results

Olsson et al. (2020), [8] has compared results with 4 datasets

	NinaPro DB-1 (200 ms) Test accuracy (%)	NinaPro DB-2 (200 ms) Test accuracy (%)	BioPatRec (150 ms) Test accuracy (%)	CapgMyo - DBa (150 ms) Test accuracy (%)
Benchmark classifier with handcrafted feature set	75.3	75.3	92.9	-
CNN by Atzori et al. [11]	66.6	60.3	-	-
CNN by Geng et al. [12]	77.8	-	-	99.5
Multistream-CNN by Wei et al.[13]	84	-	-	99.7
CNN with PCA by Zhai et al. [20]	-	78.7	-	-
CNN-RNN by Hu et al. [14]	87	82.2	94.1	99.7
Evolved CNN [8]	81	71.6	91.4	98

SPEC- CNN

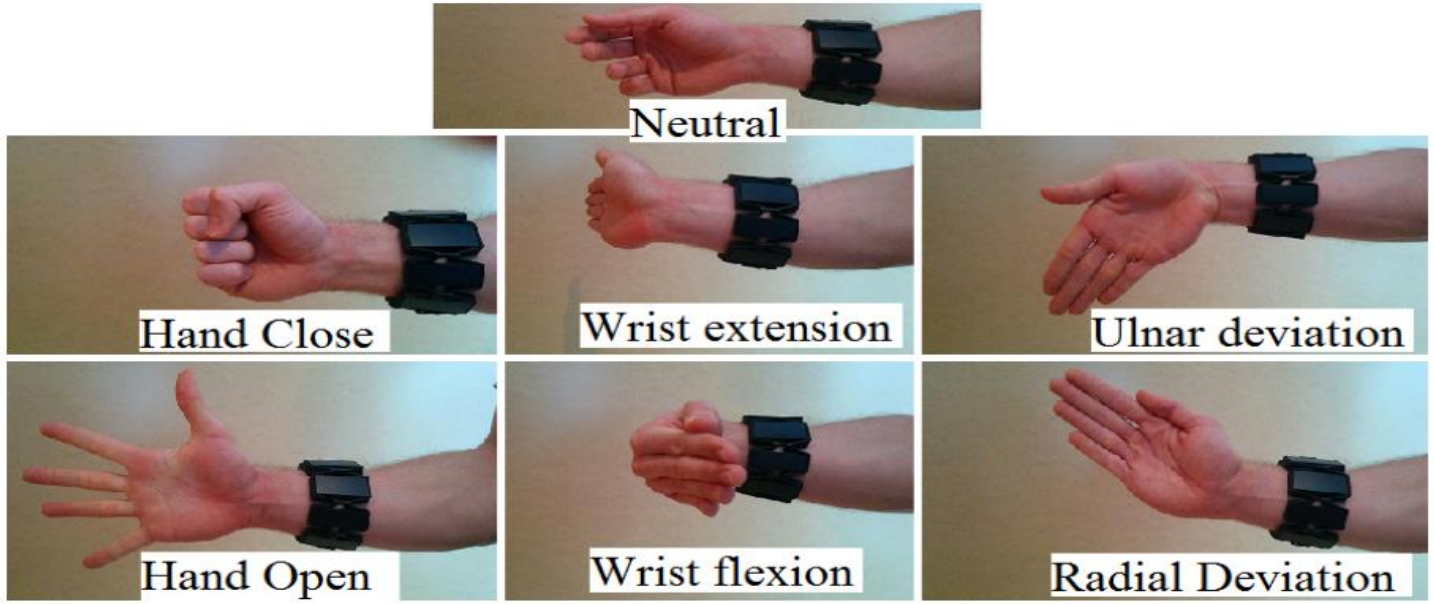
- Cote-Allard et al. (2020), [9] has worked on the following datasets
  - NinaPro DB5
  - Myo Dataset

- Considered three different modalities of input for CNN
  - raw sEMG image
  - Spectrograms
  - Continuous Wavelet Transform (CWT)

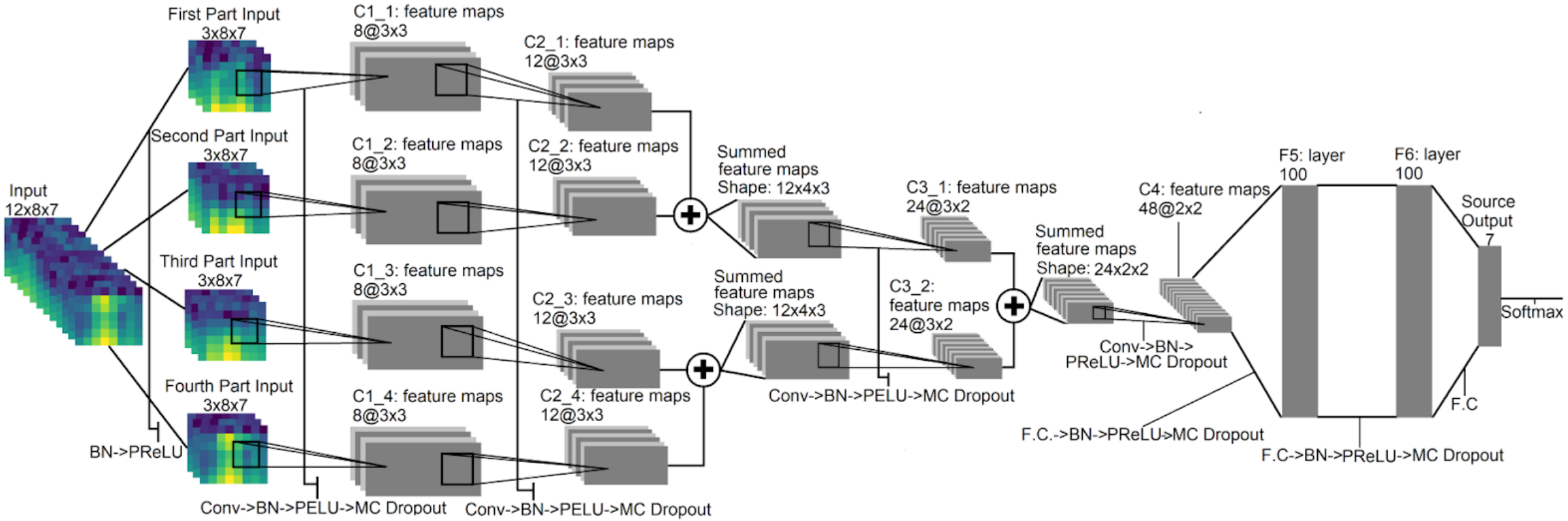
- Implemented Transfer Learning (TL) technique

SPEC- CNN on Myo Data

- Cote-Allard et al. (2020), [9] has provided the following dataset
  - Myo Dataset -7 wrist movements



SPEC- CNN


First Part Input
3x8x7
Second Part Input
3x8x7
Third Part Input
3x8x7
Fourth Part Input
3x8x7
Input
12x8x7
BN->PReLU
C1_1: feature maps
8@3x3
C1_2: feature maps
8@3x3
C1_3: feature maps
8@3x3
C1_4: feature maps
8@3x3
Conv->BN->PELU->MC Dropout
C2_1: feature maps
12@3x3
C2_2: feature maps
12@3x3
C2_3: feature maps
12@3x3
C2_4: feature maps
12@3x3
Conv->BN->PELU->MC Dropout
Summed
feature maps
Shape: 12x4x3
Summed
feature maps
Shape: 12x4x3
Conv->BN->PELU->MC Dropout
C3_1: feature maps
24@3x2
C3_2:
feature maps
24@3x2
Summed
feature maps
Shape: 24x2x2
C4: feature maps
48@2x2
Conv->BN->
PReLU->MC Dropout
F5: layer
100
F6: layer
100
Source
Output
7
Softmax
F.C
F.C.->BN->PReLU->MC Dropout
F.C->BN->PReLU->MC Dropout

SPEC- CNN- Results

➔ Cote-Allard et al. (2020), [9] results for NinaPro DB5



Hands-on (Demo)

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THANK YOU

Contact Information: anish.turlapaty@iiits.in